Chapter ②

"Magnetic effect of electric Current"

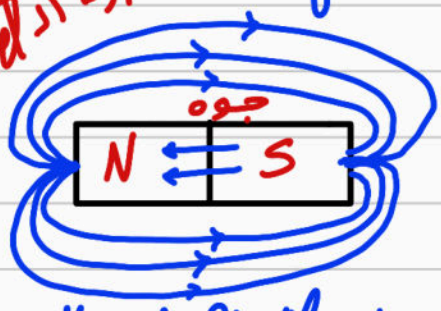
I → Mag. field

1 Basics:

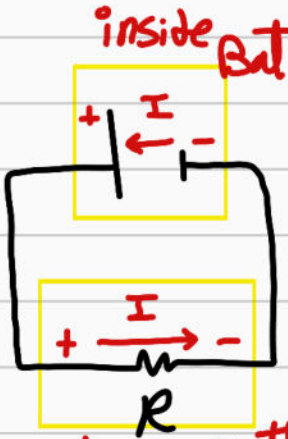
Out side the Magnet
N → S
inside the Magnet
S → N

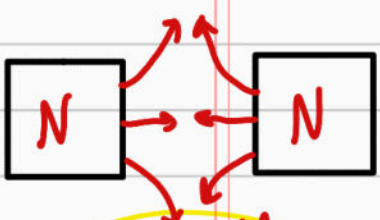
Magnet مغناطيس

Magnetic field

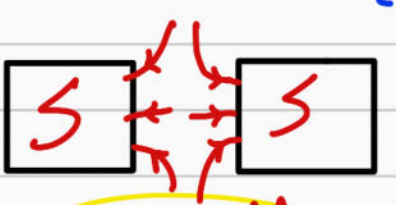


Magnetic Flux "Lines"
$[\Phi_m]$

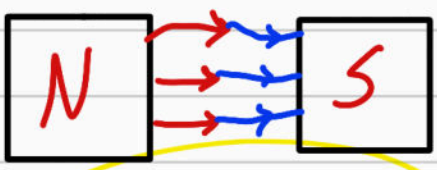




Repell

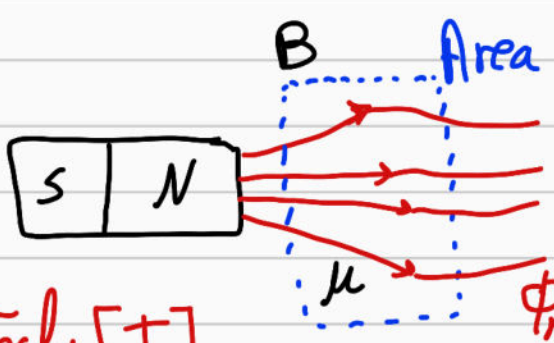


Repell



attraction

Φ_m (النماس) → Magnetic flux "Lines" (خطوط) Weber [Wb]



B (الكثافة / الشدة) → Magnetic flux density [intensity] (الشدة) Tesla [T]

μ (حلاوة / أوميجا) → Magnetic premability (ميو) [T.m/A] or [Wb/A.m]

density
Resistivity
Conductivity
premability
→ material
→ Temp.

iT is the Magnetic flux lines Passing normally through

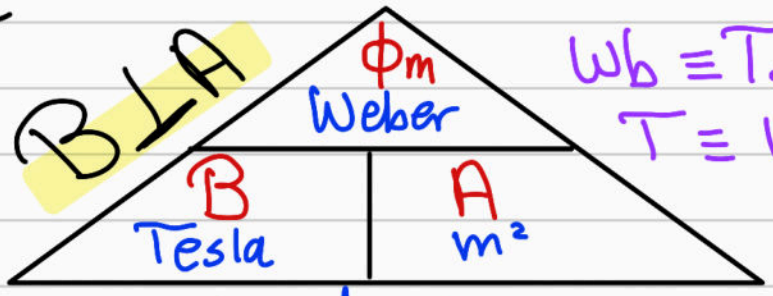
Certain Area
"Φ_m"

Unit Area
"B"





if the Magnetic flux density is normal To the Area [Perpendicular]



$Wb \equiv T \cdot m^2$

$T \equiv Wb/m^2$

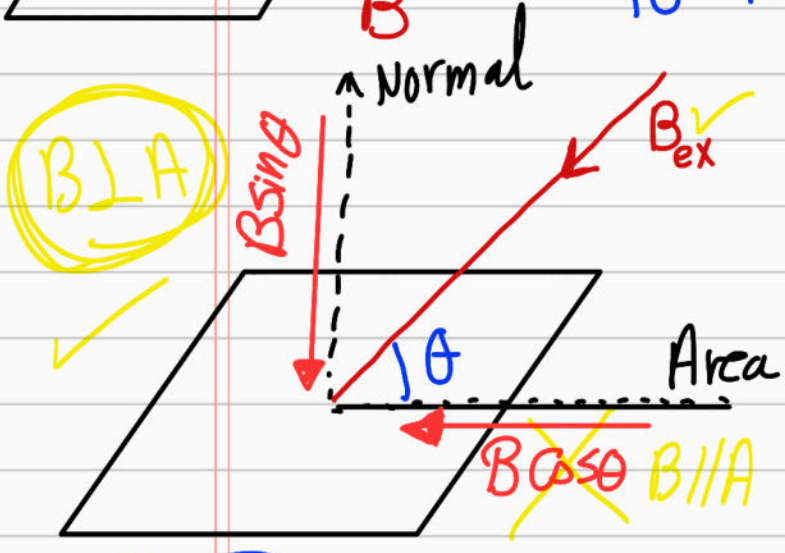
B//Area

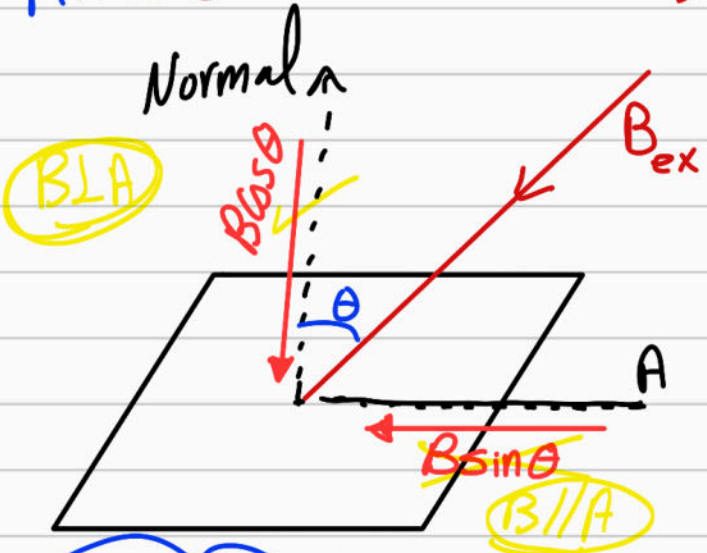


if the Magnetic flux density is

ϕ_m = Zero parallel

To the Area (No intersection)

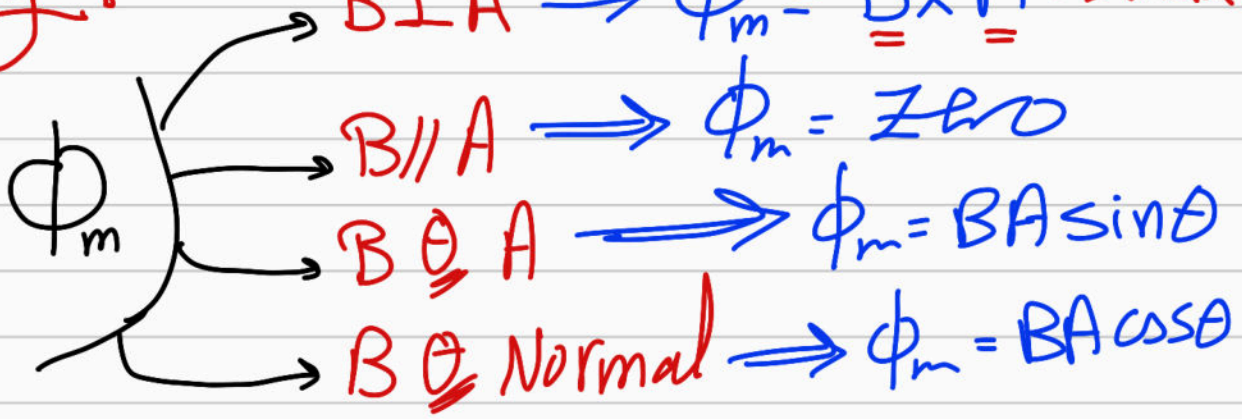




$\phi_m = BA\sin\theta$

$\phi_m = BA\cos\theta$

Summary:



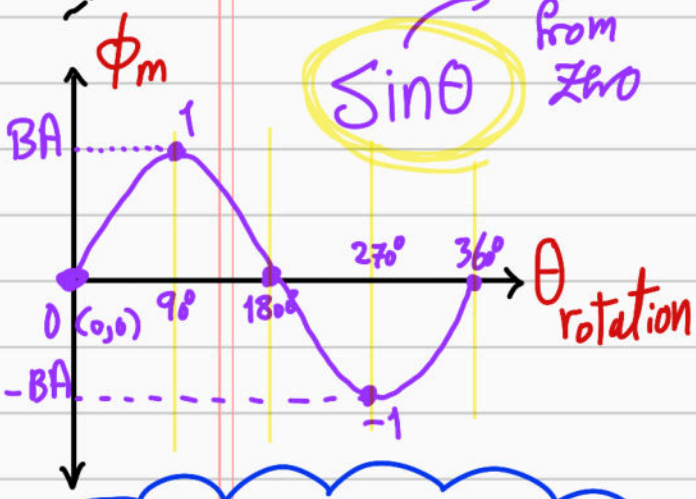
Rotation

From Zero Position

Sinθ

B // A Parallel





$$\Phi_m = BA\sin\theta$$ ✓✓ ✓ rotation

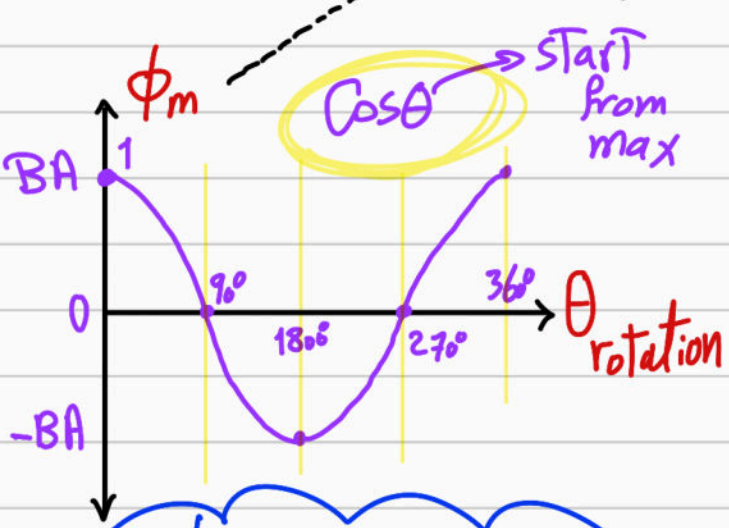
From Max Position

Perpendicular

Cosθ

B ⊥ A





$$\Phi_m = BA\cos\theta$$ rotation

Summary

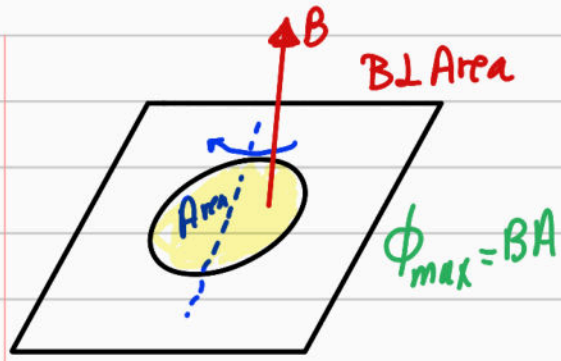
Φ_m → Mag. flux (Lines) (Wb)

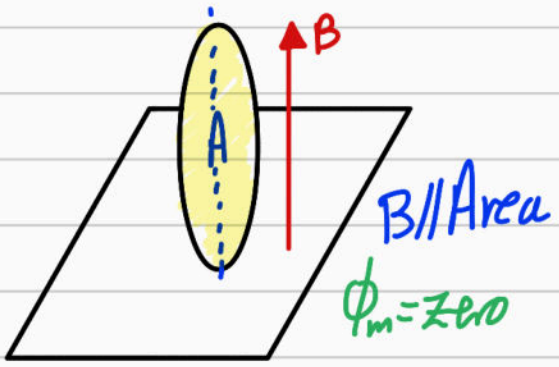
B → Mag. flux density (intensity) (T)

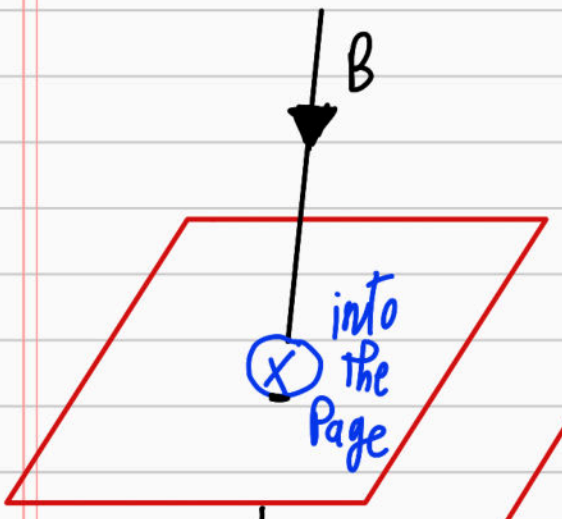
μ → Mag. Premability (T.m/A) (Wb/A.m)

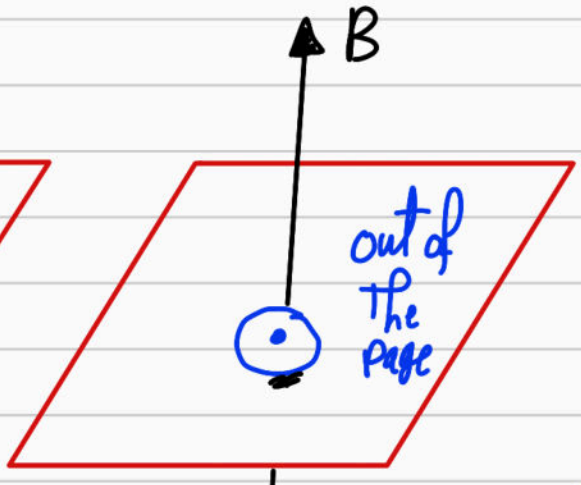
Φ_m

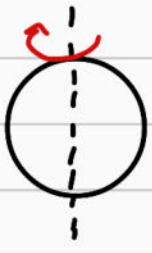
- $\Phi_m = BA$ → B ⊥ A, B // Normal
- Φ_m = Zero → B // A, B ⊥ Normal
- $\Phi_m = BA\sin\theta$ → B θ A; Rotation from Zero Position
- $\Phi_m = BA\cos\theta$ → B θ Normal; Rotation from max Pos.

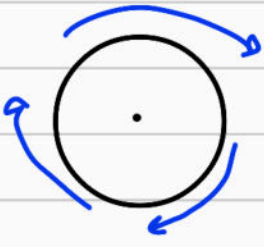




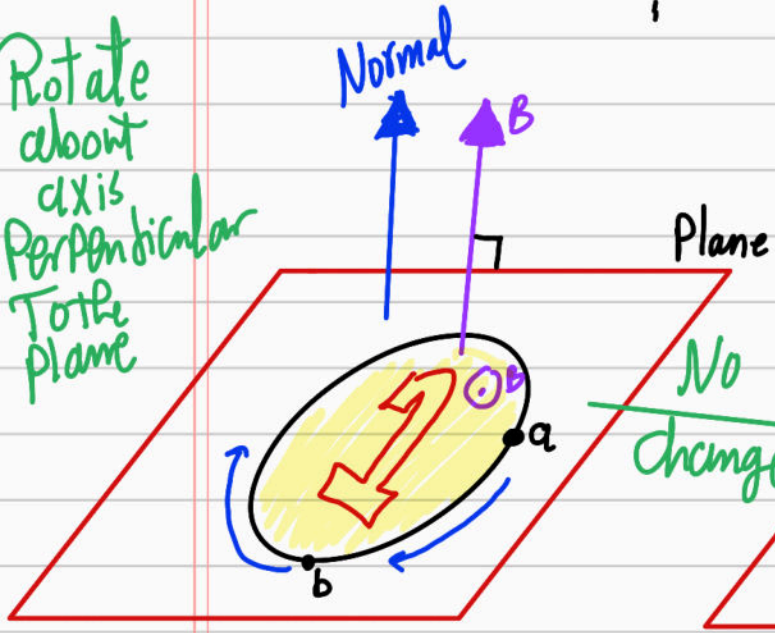




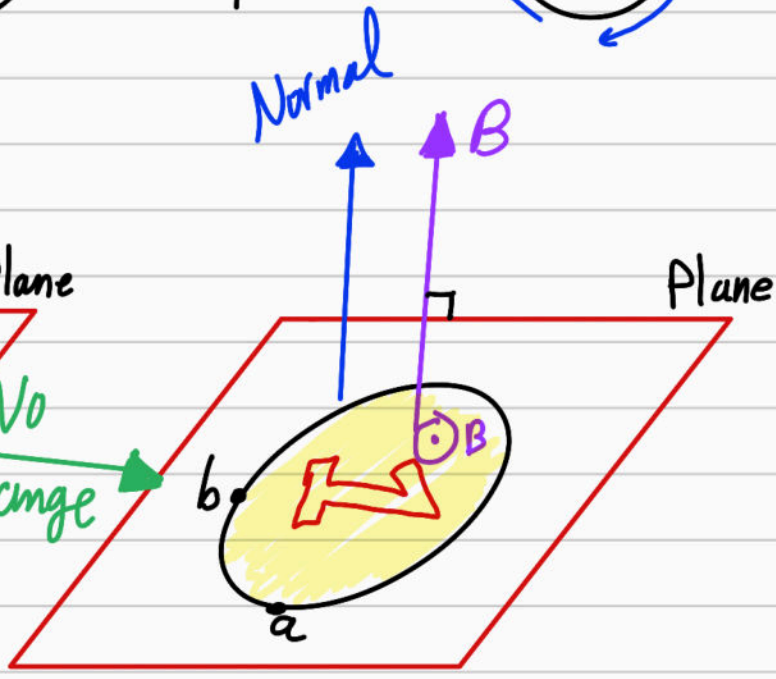




Rotate about axis perpendicular to the plane

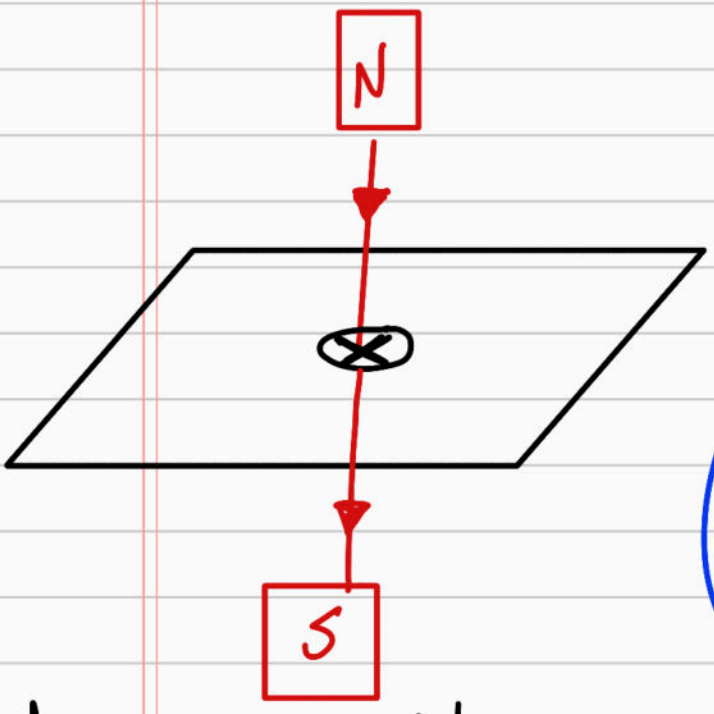


No change

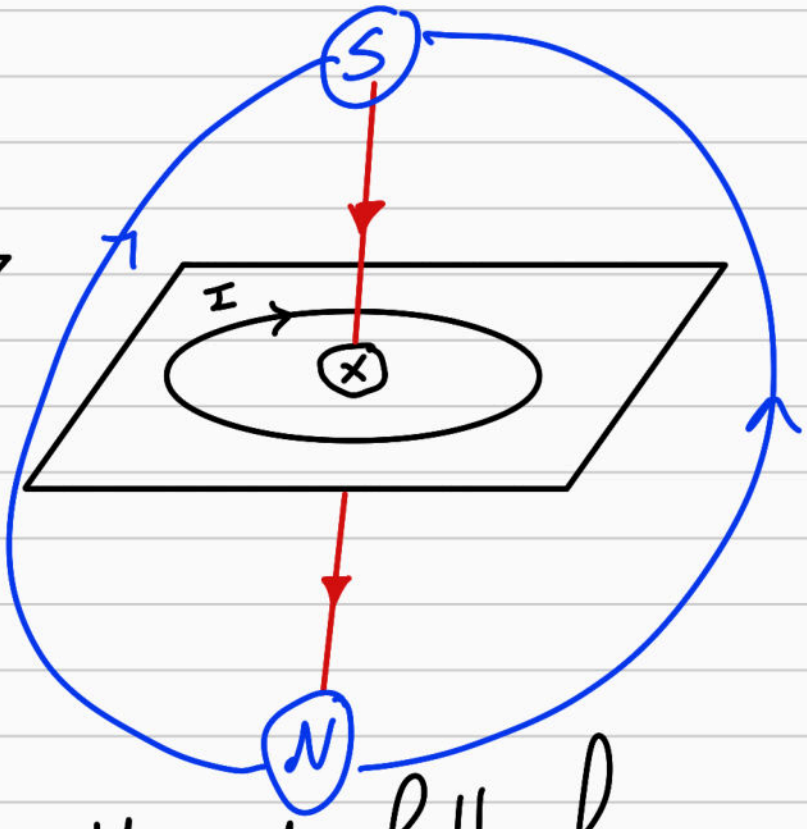


B⊥Coil plane [Area]
B// Normal

B⊥Coil plane [Area]
B// Normal



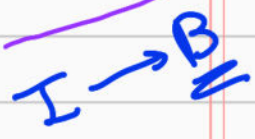
Magnetic field of external Magnet



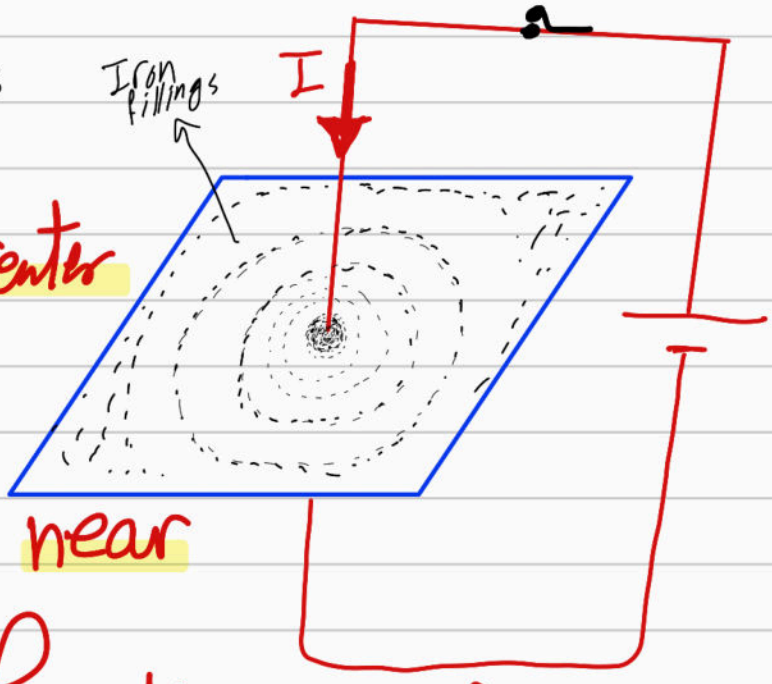
Magnetic field of Circular coil

B_{Wire}

1 Description:

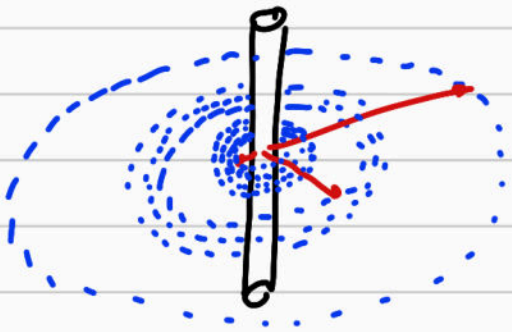


1. Concentric circles its center is the center of the Wire



2. Closer To each other near Wire $B\uparrow$

3. aWay from each other aWay from the Wire, $B\downarrow$



$e = 1.6 \times 10^{-19} C$

2 Magnitude:

$$B_{Wire} = \frac{\mu \times I}{2\pi \times d}$$

Constant

$$\mu_{air} = 4\pi \times 10^{-7} T.m/A$$



B_{Wire} → Mag. flux density due To sTraight Wire [Tesla] [T]

6 μ → Mag. Premability [T.m/A] or [Wb/A.m]

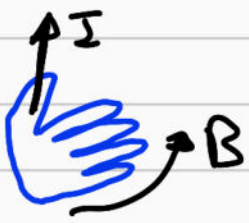
I → electric curat intensiTy passing Wire [Ampere]

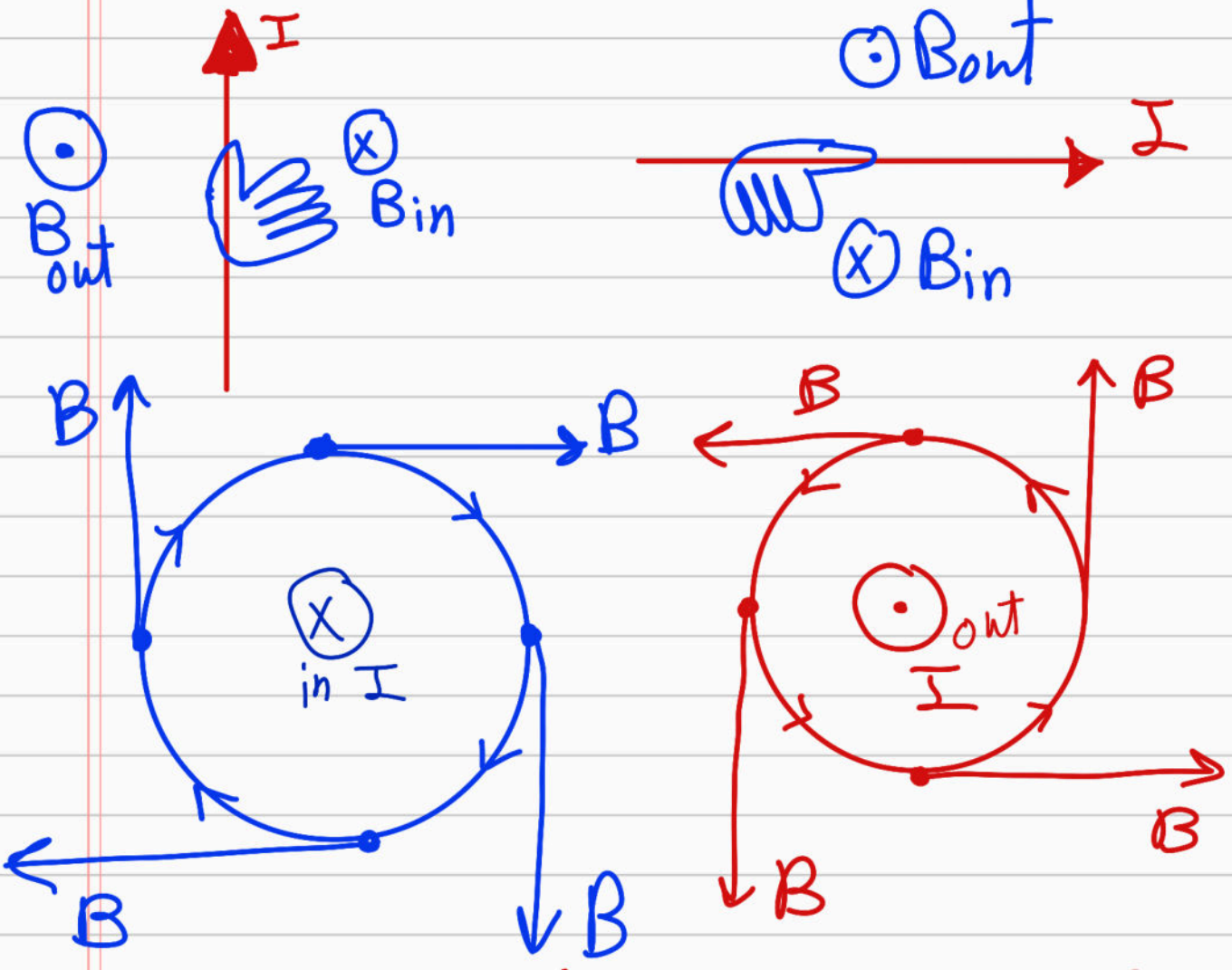
d → Normal distance between Wire & Point [meter]



$$B_{\perp} = \frac{4\pi \times 10^{-7} \times 3}{2\pi \times 20 \times 10^{-2}} = 3 \times 10^{-6} \text{ Tesla}$$

3 Direction ⇒ Ampere Right hand Rule



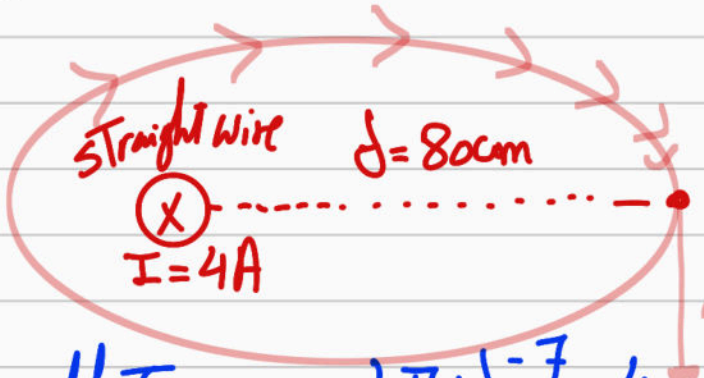


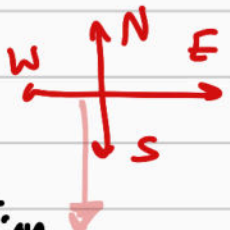
Question:

$\mu = 4\pi \times 10^{-7}$ Tim/A $e = 1.6 \times 10^{-19}$ C

⊗ in

⊙ out

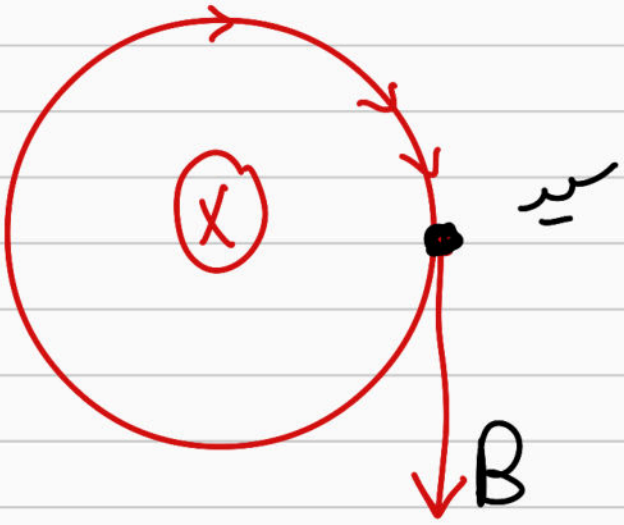




direction down

$$B_{wire} = \frac{\mu I}{2\pi d} = \frac{4\pi \times 10^{-7} \times 4}{2\pi \times 80 \times 10^{-2}} = 1 \times 10^{-6}\ T$$

Magnitude ✓

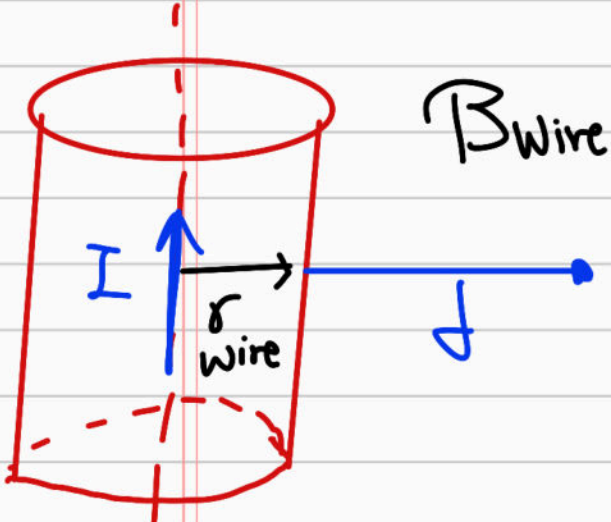


B_{wire}

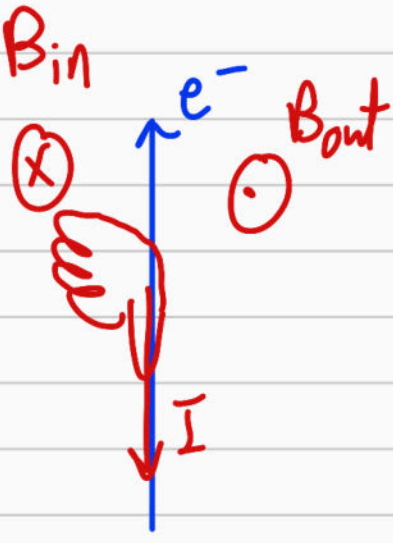
- Description
  - Concentric Circles Cater Wire
  - Closer each other near Wire
  - Away from each other away Wire
- Magnitude → $B_{wire} = \frac{\mu I}{2\pi d} = 2\times10^{-7}\left[\frac{I}{d}\right]$
  - $\mu_{air} = 4\pi\times10^{-7}\ T\cdot m/A$
- Direction → Ampere Right hand

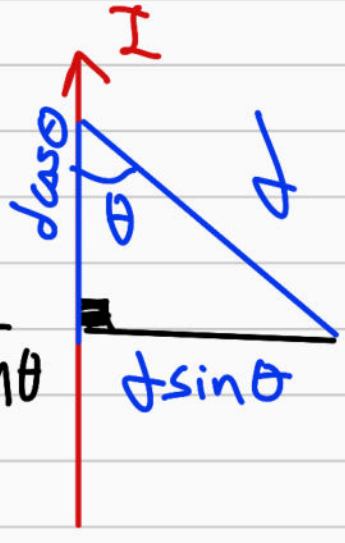


- Special Cases



$$B_{Wire} = \frac{\mu I}{2\pi[d + r_{wire}]}$$

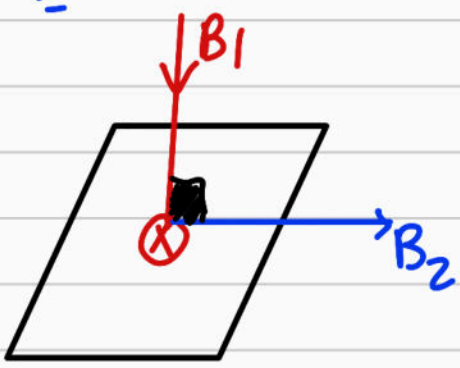


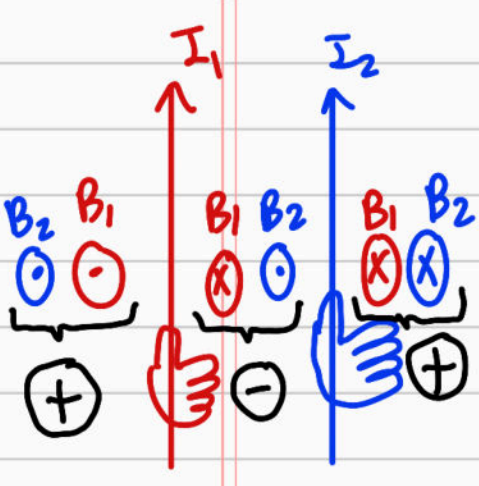


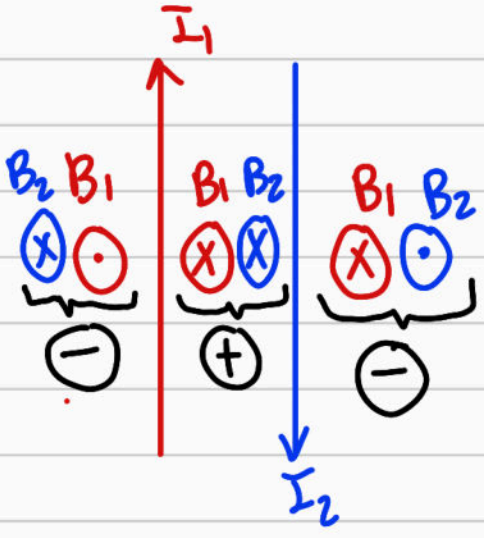
$$B_{Wire} = \frac{\mu I}{2\pi d\sin\theta}$$

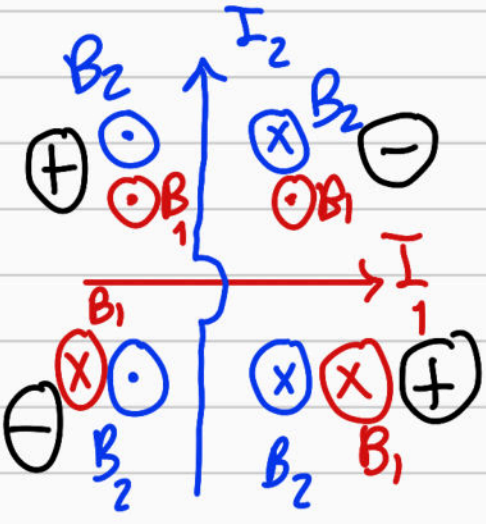
B total

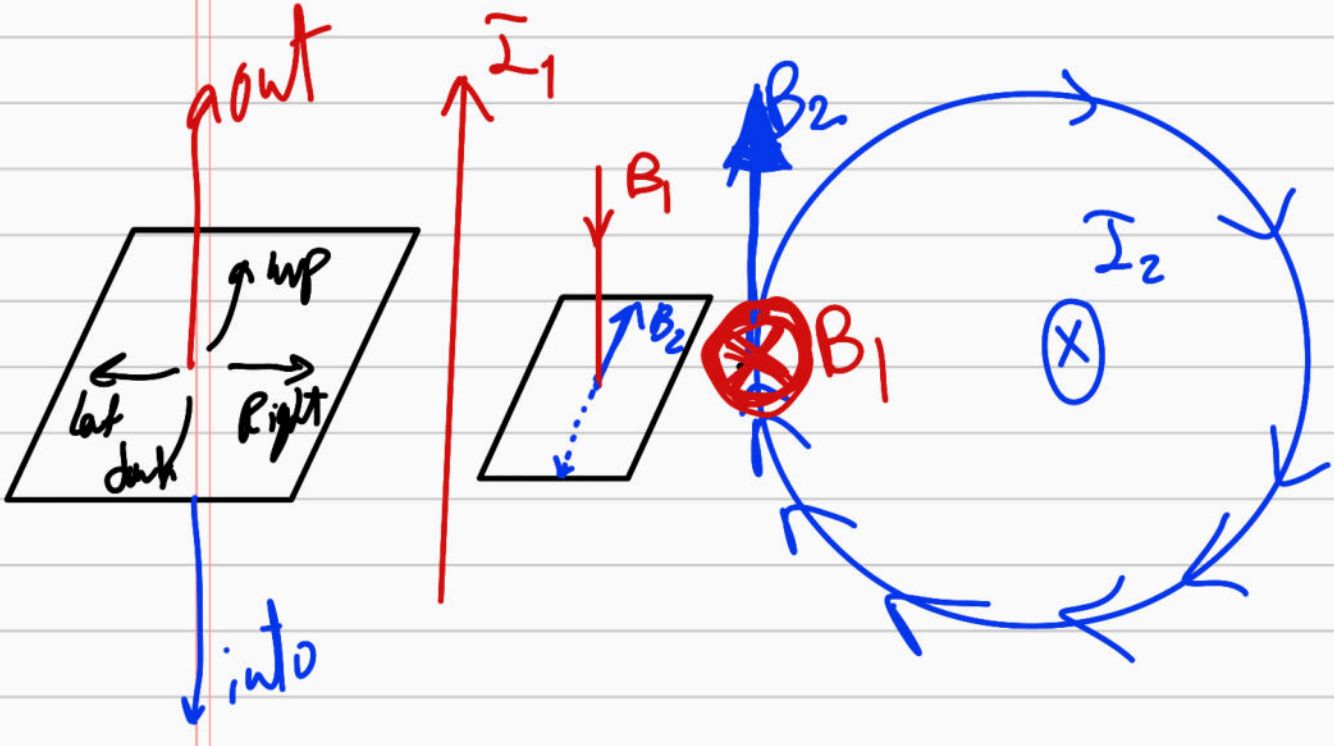
- Same → $B_t = B_1 + B_2$
- opposite → $B_t = B_{\text{الكبير}} - B_{\text{الصغير}}$
- Perpendicular → $B_t = \sqrt{B_1^2 + B_2^2}$









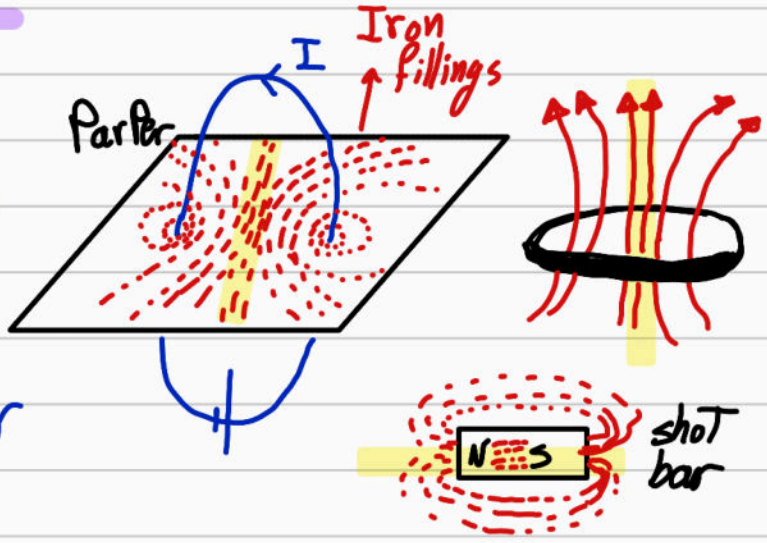


Lesson ②

Part ① Circular coil:

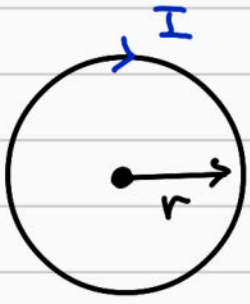
① Description:

1. Uniform* magnetic field at the center* of the coil.
2. similar To Short bar Magnent



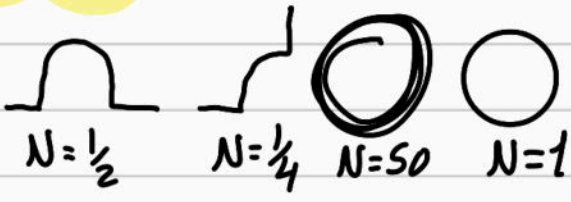
② Magnitude:

$$B_{coil} = \frac{\mu N I}{2r}$$



$\mu_{air} = 4\pi \times 10^{-7}$ T.m/A

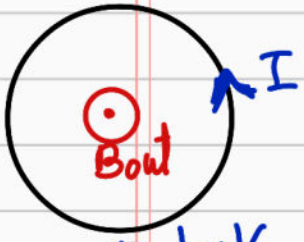
N → no. of Turns of the coil



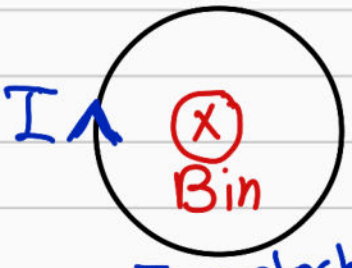
③ Direction: Right hand screw Rule



2D

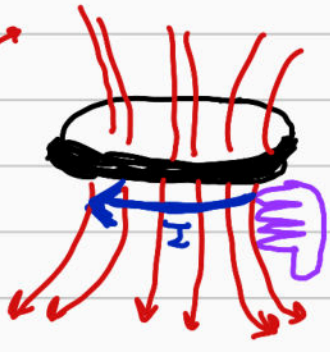


I → anti clock
B → out



I → clock wise
B → in





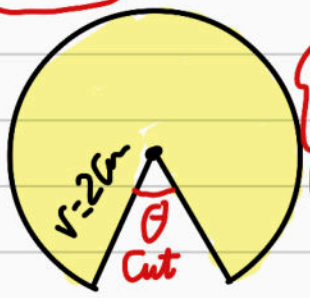
4 Special Cases:

1. Loop / Frame / Ring ⇒ N = 1

2. Sector (N < 1)

$N = \frac{\theta^\circ}{360^\circ}$

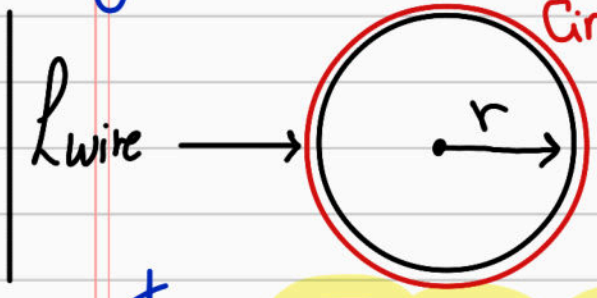




$N = \frac{360^\circ - \theta^\circ_{cut}}{360^\circ}$

$1 - \frac{1}{6} = \frac{5}{6}$

3. Length Wire ⟶ Circular Coil



Circumference

Rule

$N_{coil} = \frac{L_{wire}}{2\pi r_{coil}}$

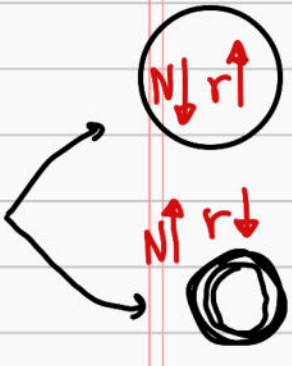
Ratio

$\frac{N_1}{N_2} = \frac{L_1}{L_2} \times \frac{r_2}{r_1}$

$N_{shape} = \frac{L_{wire}}{\text{Perimeter of shape}}$

4. Recoiled* / Reformed** / Coiled again*

⟶ Same Length ⟵



$\frac{N_1}{N_2} = \frac{r_2}{r_1}$

r → بیتغییر یا میرنا

$B_{coil} = \frac{MNI}{2r} \rightarrow \frac{B_1}{B_2} = \frac{N_1}{N_2} \times \frac{I_1}{I_2} \times \frac{r_2}{r_1}$

$\frac{N_1}{N_2}$

$\frac{B_1}{B_2} = \left(\frac{N_1}{N_2}\right)^2 \times \frac{I_1}{I_2} = \frac{I_1}{I_2} \times \left(\frac{r_2}{r_1}\right)^2$

5. Cut / Add

r → Constant

$N = \frac{L}{2\pi r} \rightarrow r = \frac{L}{2\pi N}$

$B_{Coil} = \frac{\mu N I}{2r} \rightarrow \frac{B_1}{B_2} = \frac{N_1}{N_2} \times \frac{I_1}{I_2}$

Same Current → $\frac{B_1}{B_2} = \frac{N_1}{N_2}$

$I = \frac{V}{R} \rightarrow \frac{B_1}{B_2} = \frac{N_1}{N_2} \times \frac{V_1}{V_2} \times \frac{R_2}{R_1}$

$R = \frac{\rho l}{A} \rightarrow \frac{B_1}{B_2} = \frac{N_1}{N_2} \times \frac{V_1}{V_2} \times \frac{l_2}{l_1}$

$L \rightarrow L = N \times 2\pi r$

$\frac{B_1}{B_2} = \frac{N_1}{N_2} \times \frac{V_1}{V_2} \times \frac{N_2}{N_1}$

$\frac{B_1}{B_2} = \frac{V_1}{V_2}$ → Same Source (V) → $\frac{B_1}{B_2} = \frac{1}{1}$

$B_1 = B_2$

Cut / Add

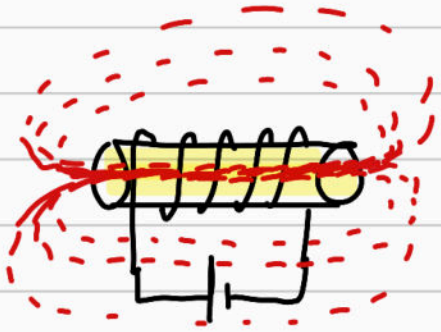
B_{Coil}

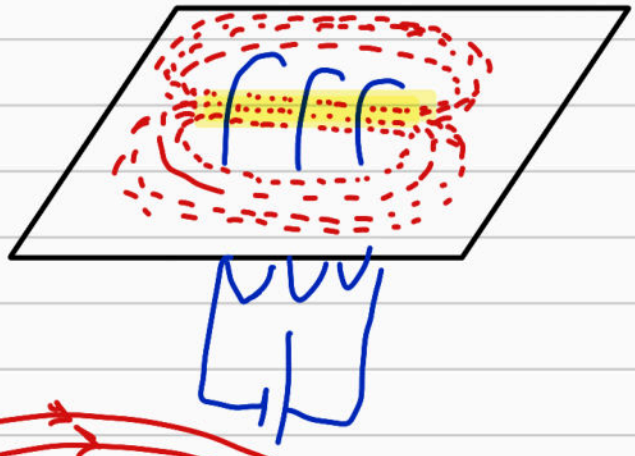
- Generally $\frac{B_1}{B_2} = \frac{N_1}{N_2} \times \frac{I_1}{I_2}$
- Same Current $\frac{B_1}{B_2} = \frac{N_1}{N_2}$
- Same source $B_1 = B_2$
- different Source $\frac{B_1}{B_2} = \frac{V_1}{V_2}$

99%

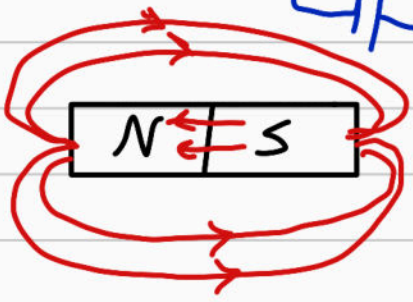
$I \propto \frac{V}{R} \propto \frac{1}{\rho l} \propto \frac{A}{N \times 2\pi r}$

$B_{solenoid}$

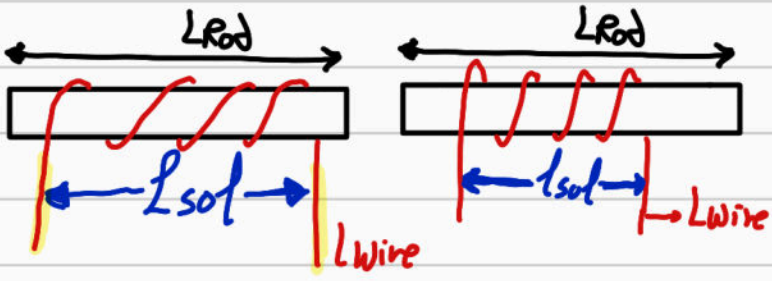




1. Uniform magnetic field at the axis of solenoid.
2. Similar long bar Magnet



$$B_{sol.} = \frac{\mu N I}{L_{sol}}$$

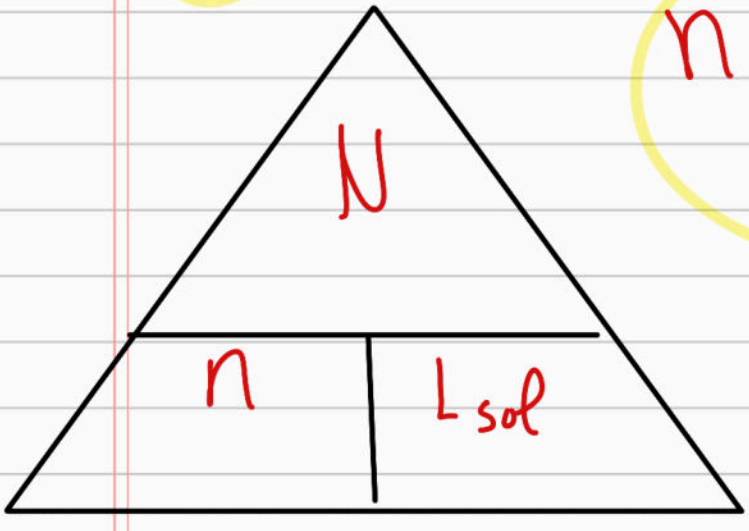


$$B_{sol.} = \mu n I$$

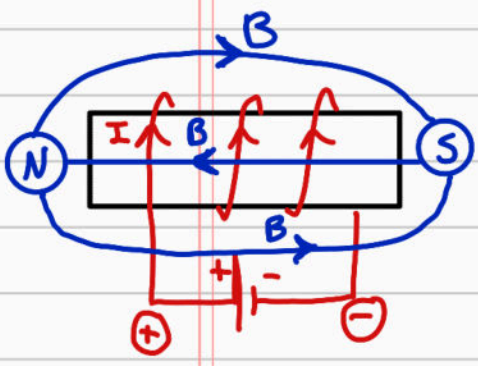
$N \rightarrow$ no. of Turns

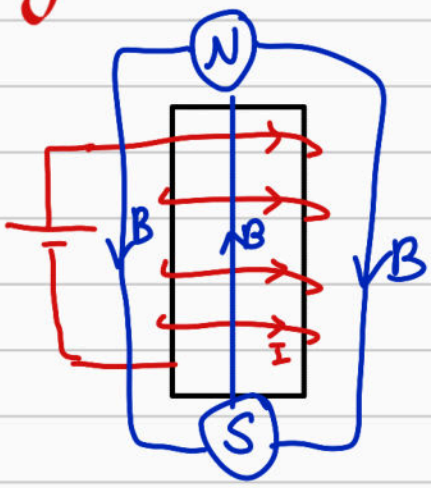
$n \rightarrow$ no. of Turns per unit Length

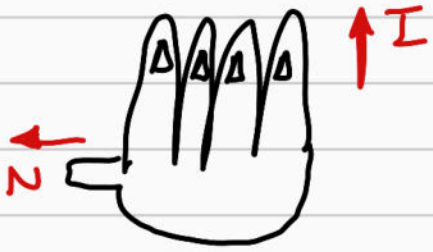
$$n = \frac{N}{L_{sol}}$$



Direction: Right hand Rule by Ampere





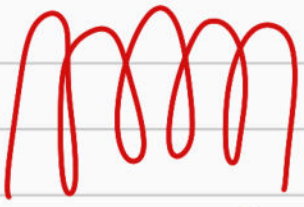


$$\frac{B_{coil}}{B_{sol.}} = \frac{\frac{\mu N \Delta I}{2r}}{\frac{\mu N I}{l_s}}$$

Special Cases:

1. Coil $\underset{\text{compress}}{\overset{\text{stretch}}{\rightleftarrows}}$ Solenoid





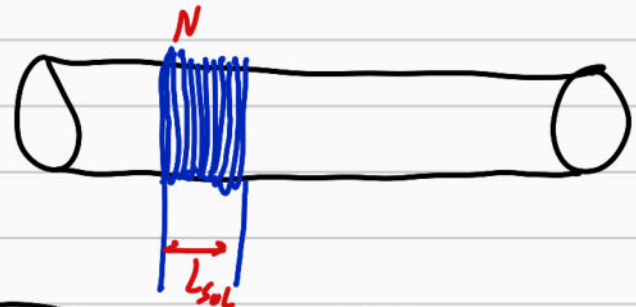
$N_{coil} = N_{solenoid}$

same I → $\frac{B_{coil}}{B_{sol.}} = \frac{l_{sol}}{2r_{coil}}$

$$\frac{B_{coil}}{B_{sol.}} = \frac{I_{coil}}{I_{sol.}} \times \frac{l_{sol}}{2r_{coil}}$$

2. Turns are Touching / adjacent / Tangent

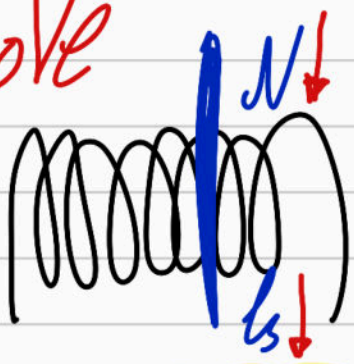
$L_{sol} = N \times D_{wire}$



$$B_{sol.} = \frac{\mu N I}{N \times D_{wire}}$$

$$B_{sol.} = \frac{\mu I}{D_{wire}}$$

Cut / Add / Remove



$n = \frac{N\downarrow}{L_s\downarrow}$ (L const.)

$B_{sol.} = \mu n I \rightarrow \frac{B_1}{B_2} = \frac{I_1}{I_2} \xrightarrow[I]{\text{Same}} B_1 = B_2$

$\frac{B_1}{B_2} = \frac{V_1}{V_2} \times \frac{R_2}{R_1}$

$I = \frac{V}{R} \rightarrow R = \frac{\rho l}{A} \rightarrow l = N \times 2\pi r$

$= \frac{V_1}{V_2} \times \frac{L_2}{L_1}$

$\frac{B_1}{B_2} = \frac{V_1}{V_2} \times \frac{N_2}{N_1} \xrightarrow[\text{Sowce}]{\text{Same}} \frac{B_1}{B_2} = \frac{N_2}{N_1}$ (V)

Cut / Remove Add	Circular coil	Solenoid
Rule	$B_c = \frac{\mu N I}{2r}$	$B_s = \mu n I$
Constant	μ / r $r \rightarrow$ const.	μ / n $n = \frac{N\downarrow}{L_s\downarrow} \rightarrow$ const.
Same Current	$\frac{B_1}{B_2} = \frac{N_1}{N_2}$	$B_1 = B_2$
Same Source	$B_1 = B_2$	$\frac{B_1}{B_2} = \frac{N_2}{N_1}$
diffat Source	$\frac{B_1}{B_2} = \frac{V_1}{V_2}$	$\frac{B_1}{B_2} = \frac{V_1}{V_2} \times \frac{N_2}{N_1}$

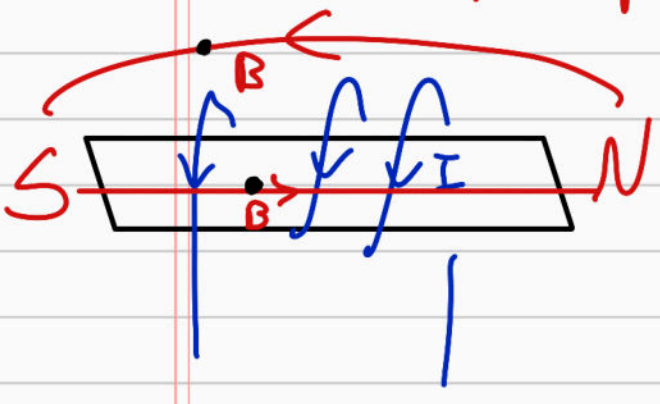
Solenoid → $B_S = \frac{\mu N I}{L_s}$

→ $B_S = \mu n I$

→ $n = \frac{N}{L_s}$

Tangat → $L_s = N \times D_{Wire}$

→ Ampare Right

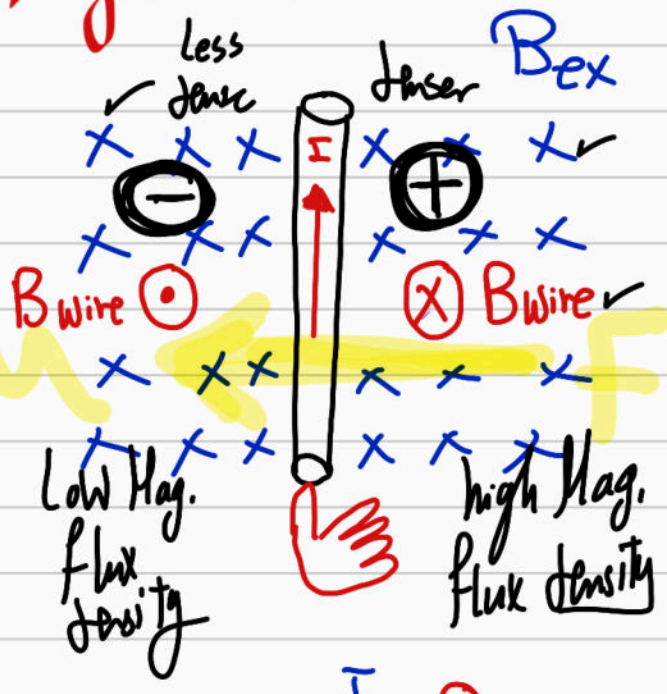




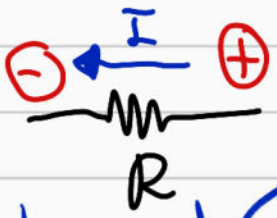
Chapter ② Lesson ③ : Magnetic force

① Concept:

The force is Genrated as Result of the Magnetic flux dasity differnce between the Two Regions around the Wire.



high → less

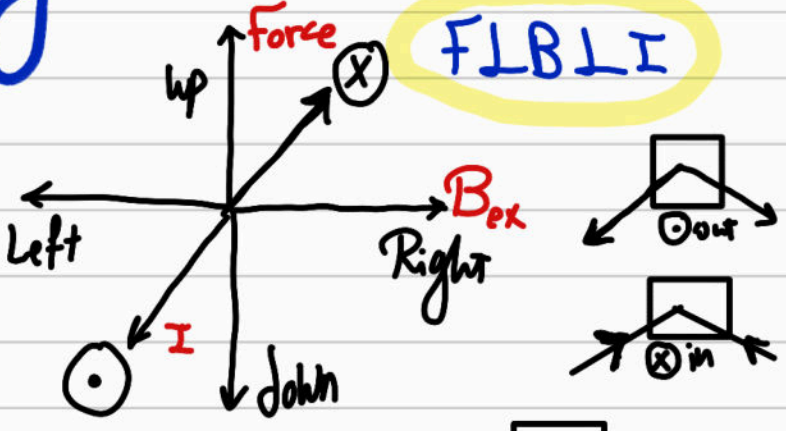


② Direction: Fleming Left hand Rule

Thumb ⟶ Force

Pointer [fore] ⟶ B_{ex}

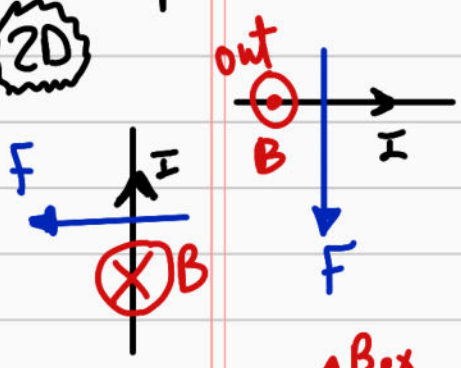
Rest of fingers ⟶ I

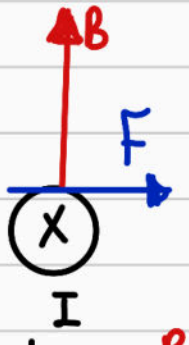


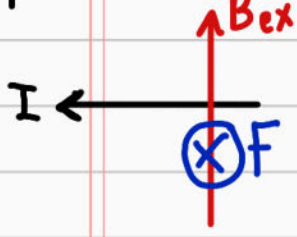
F⊥B⊥I

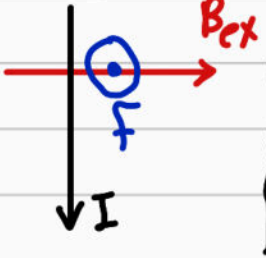
examples:

2D



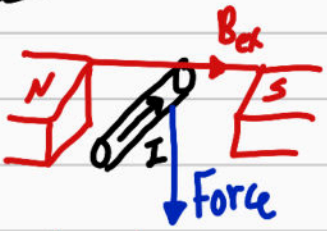


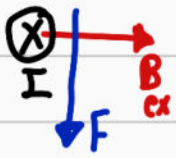


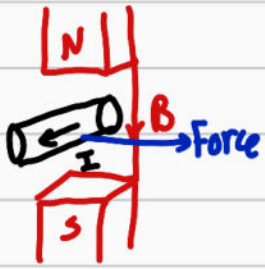


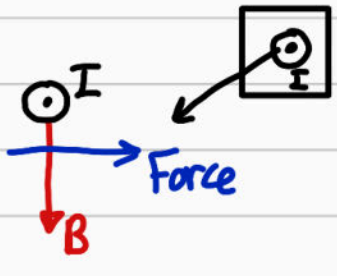
3D









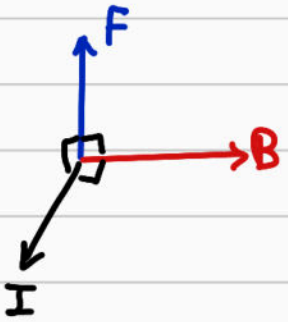


3 Magnitude:

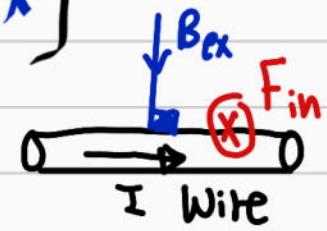
$F \propto L$
$F \propto I$
$F \propto B_{ex}$

$F \propto L I B_{ex}$

$F = L I B_{ex} \rightarrow B_{ex} \perp I$

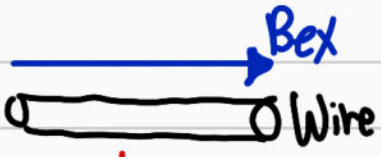


1 $B_{ex} \perp$ Wire



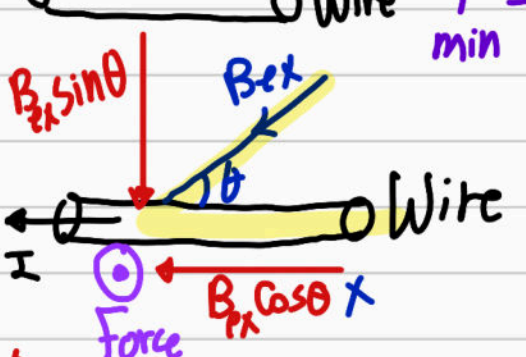
$F_{max} = L I B_{ex}$

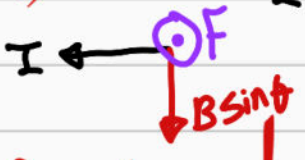
2 $B_{ex} \parallel$ Wire

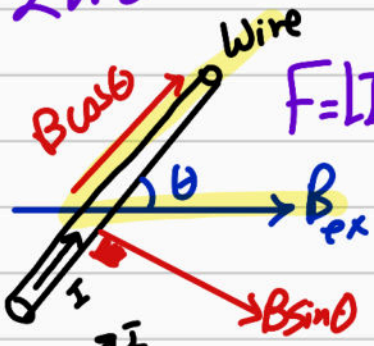


$F_{min} = Zero$

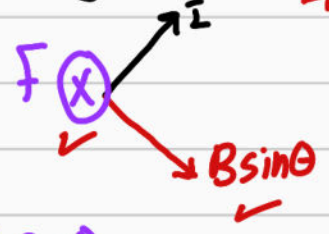
3 B_{ex} θ Wire

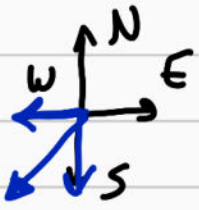




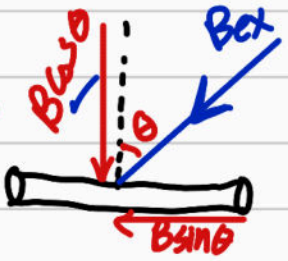


$F = L I B_{ex} \sin\theta$



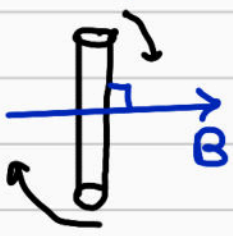


4 B_{ex} θ Normal



$F = L I B_{ex} \cos\theta$

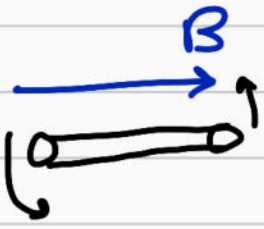
5 Rotation from $B_{ex} \perp$ Wire

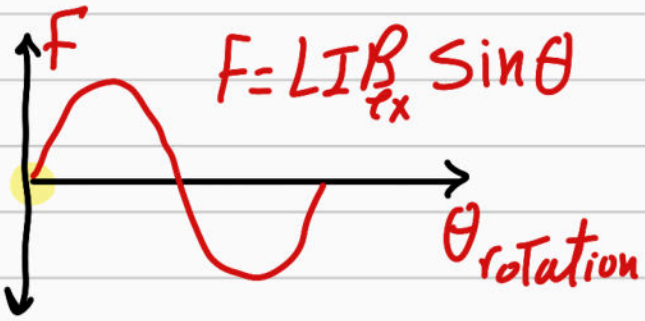




$F = L I B_{ex} \cos\theta$

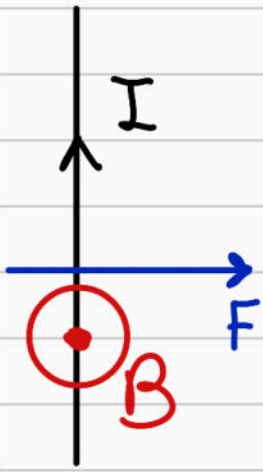
6 Rotation from $B_{ex} \parallel$ Wire

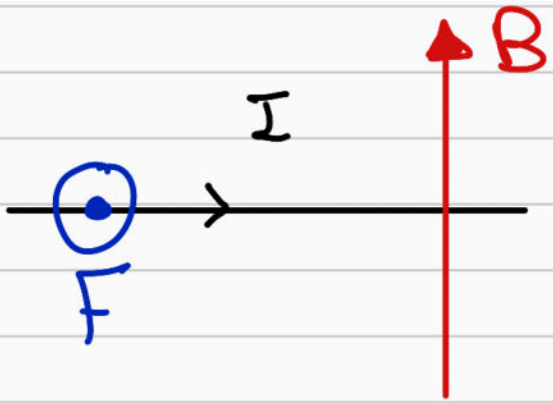


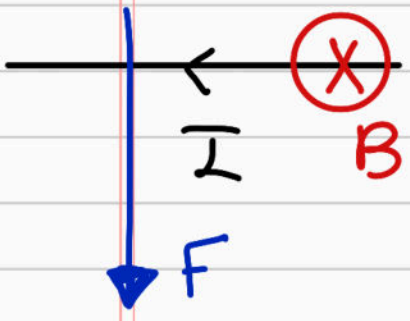


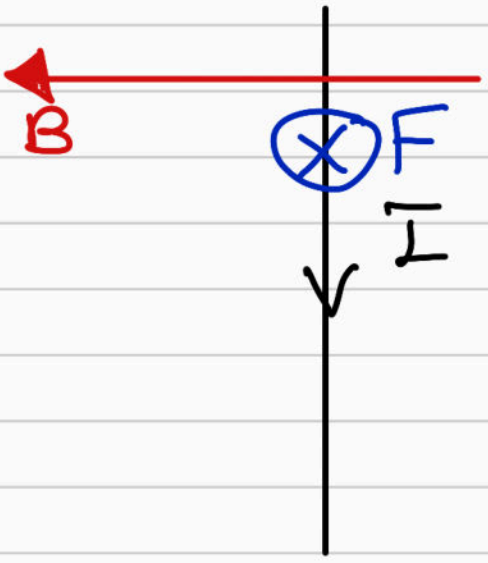
$F = L I B_{ex} \sin\theta$

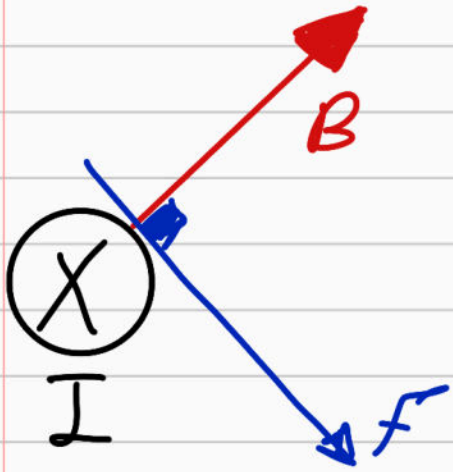
Direction

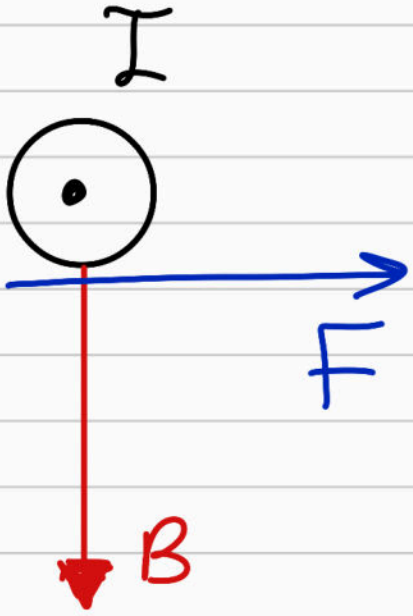




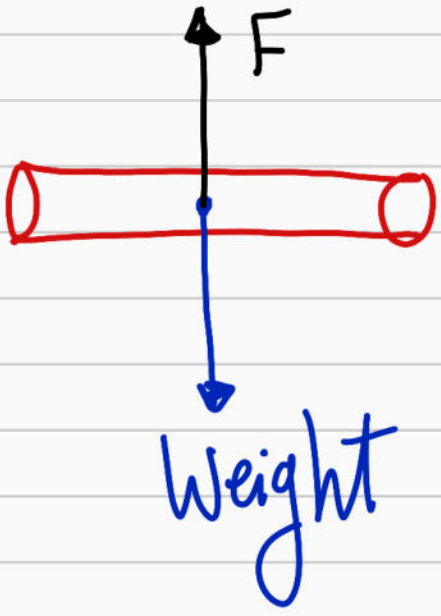








Equillibrim $\Sigma F = 0$



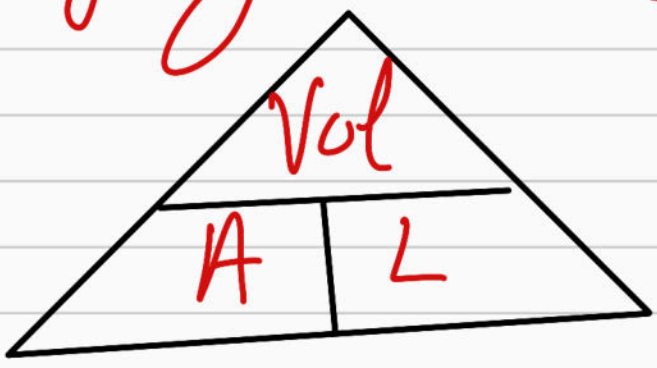
$F = Weight$

$$LIB_{ex} = m \times g$$

$m \rightarrow$ mass [Kg]

$g \rightarrow$ accelration due To gravity $\rightarrow [9.8 m/s^2]$





$$LIB = \rho Vol \times g$$

$$LIB = \rho A \cancel{L} g$$

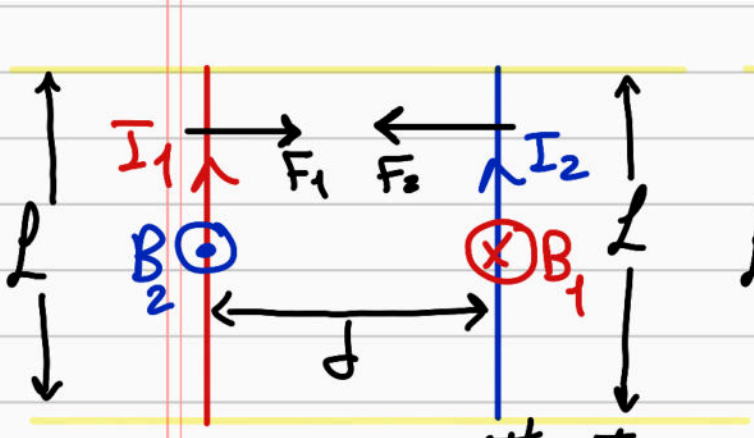
$$IB_{ex} = \rho A g$$

examples:

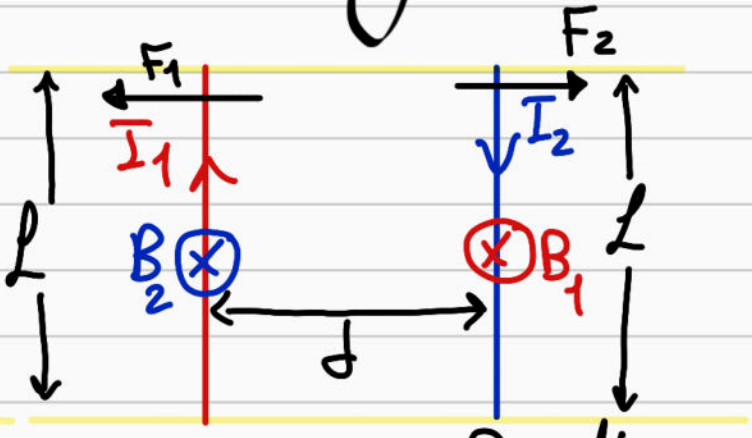
1. Without any external support
2. No Tension in Wires
3. Suspension Without any support
4. hanged Without any extend support
5. Balanced
6. static

3 Mutual Force for Two Wires only

$F_1 = F_2$



Two Wires attract each other



Two Wires Repell each other

$$F_1 = L I_1 B_2 = L I_1 \times \frac{\mu I_2}{2\pi d} = \frac{\mu I_1 I_2 L}{2\pi d}$$

$$F_2 = L I_2 B_1 = L I_2 \times \frac{\mu I_1}{2\pi d} = \frac{\mu I_1 I_2 L}{2\pi d}$$

$$F_1 = F_2 = F_{Mutual} = \frac{4\pi \times 10^{-7} \mu I_1 I_2 L_{common}}{2\pi d}$$

$$F_{Mutual} = 2 \times 10^{-7} \left[\frac{I_1 I_2 L_{comm}}{d} \right]$$



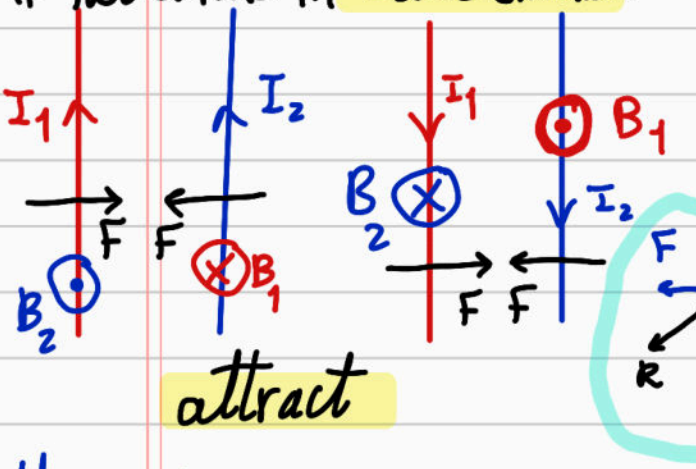
Mutual Force Summary for Two Wires only

"Without any external Magnetic field"



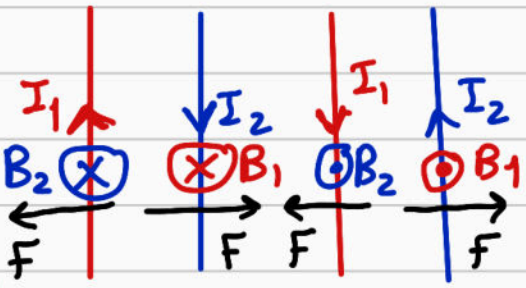
#type of Mutual Force:

if Two Currents in same direction ✓



attract

if Two Currents in opposite directions ✓



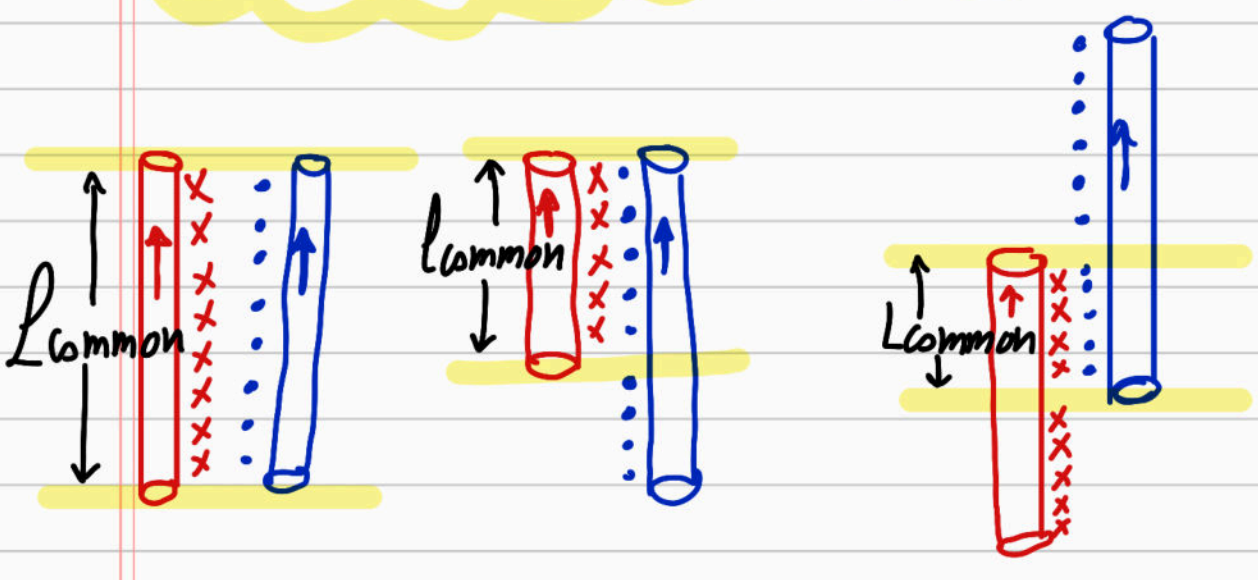
Repell

#Magnitude:

$$F_1 = F_2 = F_{Mutual} = \frac{\mu I_1 I_2 \ell_{common}}{2\pi d}$$

$\mu = 4\pi \times 10^{-7}\ T\cdot m/A$

$$F_{Mutual} = 2\times 10^{-7}\left[\frac{I_1 I_2 \ell_{common}}{d}\right]$$



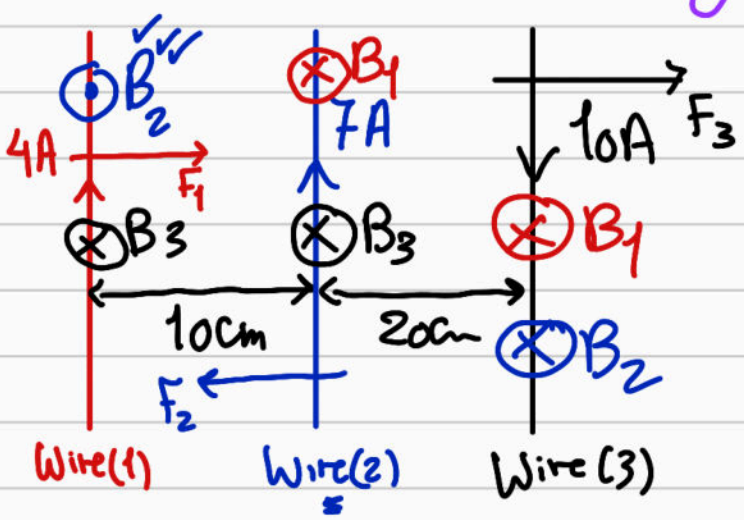
Three or more Wires Problem: all Wires are 1 meter long

$F_1 = L \underset{\text{common}}{I_1} B_{ex}$ (B_{ex}: B_2, B_3)

$$F_1 = 1 \times 4 \times 2\times10^{-7}\left[\frac{7}{0.1} - \frac{10}{0.3}\right]$$

$$F_2 = 1 \times 7 \times 2\times10^{-7}\left[\frac{4}{0.1} + \frac{10}{0.2}\right]$$

$$F_3 = 1 \times 10 \times 2\times10^{-7}\left[\frac{7}{0.2} + \frac{4}{0.3}\right]$$

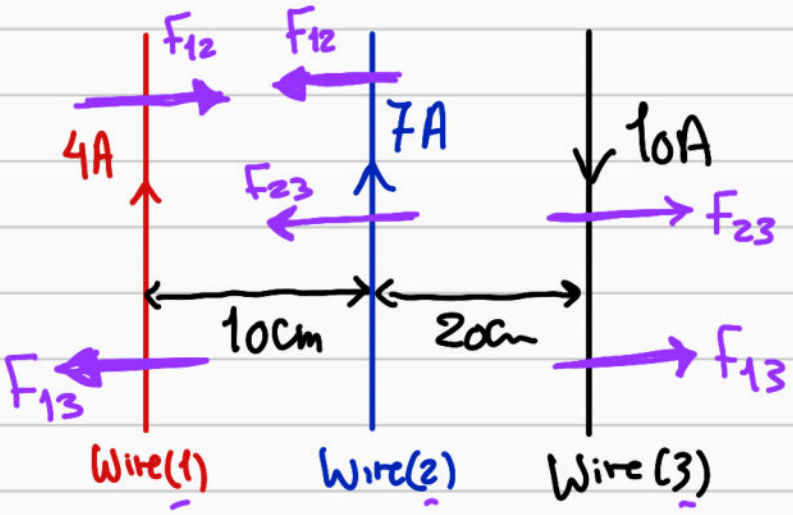


* Another solution

$$F_{12} = 2\times10^{-7}\left[\frac{4\times7\times1}{0.1}\right]$$

$$F_{23} = 2\times10^{-7}\left[\frac{7\times10\times1}{0.2}\right]$$

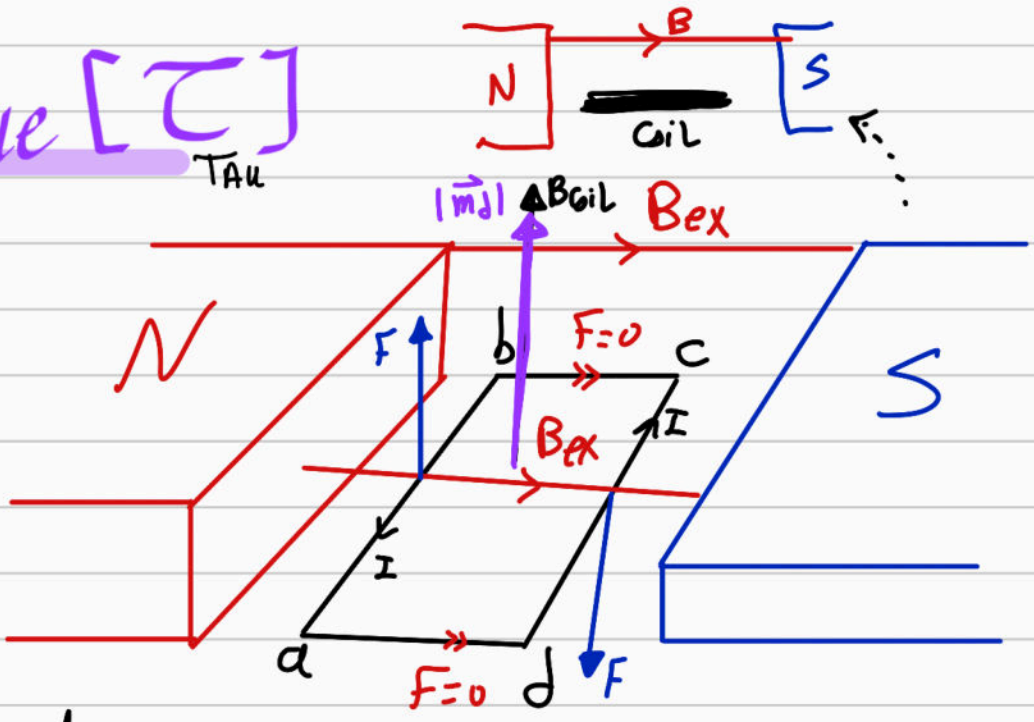
$$F_{13} = 2\times10^{-7}\left[\frac{4\times10\times1}{0.3}\right]$$



$$F_1 = F_{12} - F_{13} \qquad F_2 = F_{12} + F_{23} \qquad F_3 = F_{13} + F_{23}$$

Magnetic Torque [τ]

TAU



$$\tau = F_{ab} \times d_{bc}$$

$$= L_{ab} I B_{ex} \times L_{bc}$$

$$= B_{ex} I A \longrightarrow \text{single Turn}$$

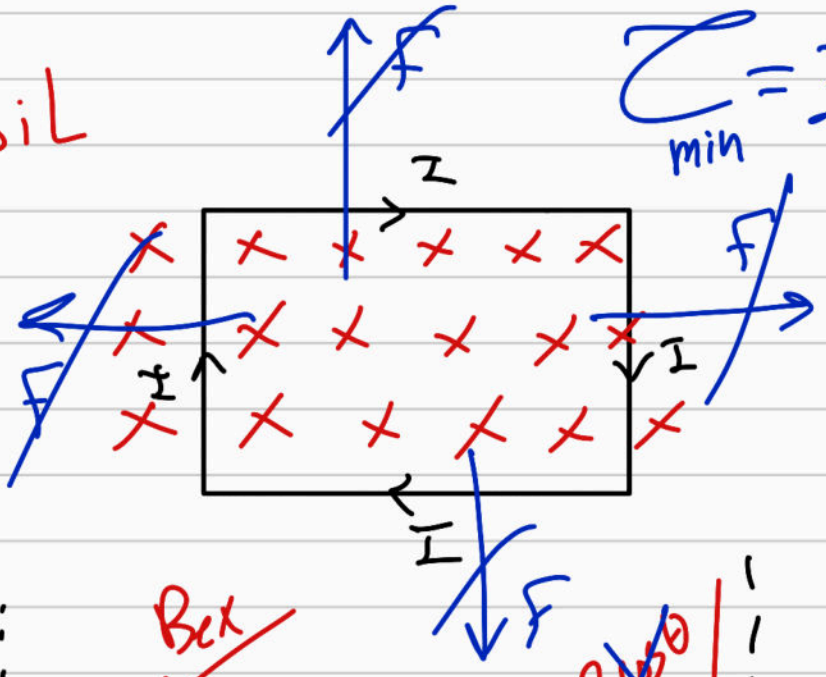
$$= B_{ex} I A N \longrightarrow \square \text{ N Turns}$$

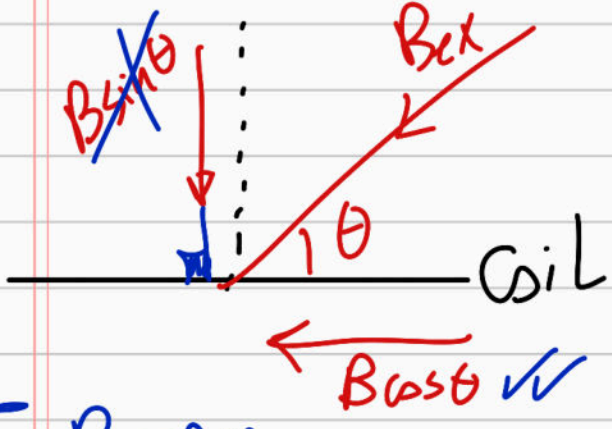
$B_{ex} \parallel$ Coil

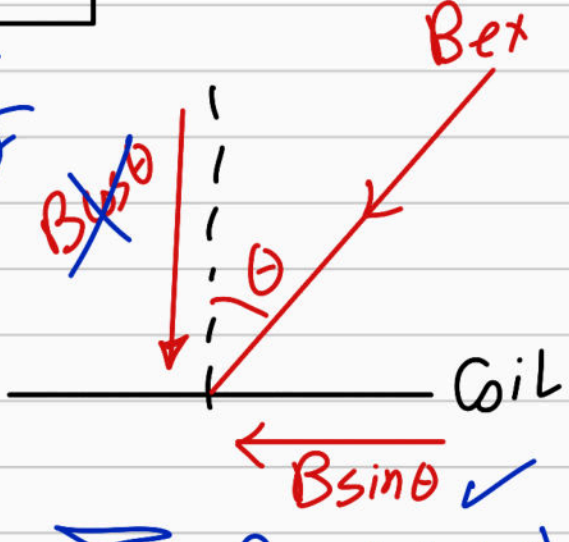
$\tau_{max} = B_{ex} I A N$

$B_{ex} \perp$ Coil

$\tau_{min} =$ Zero







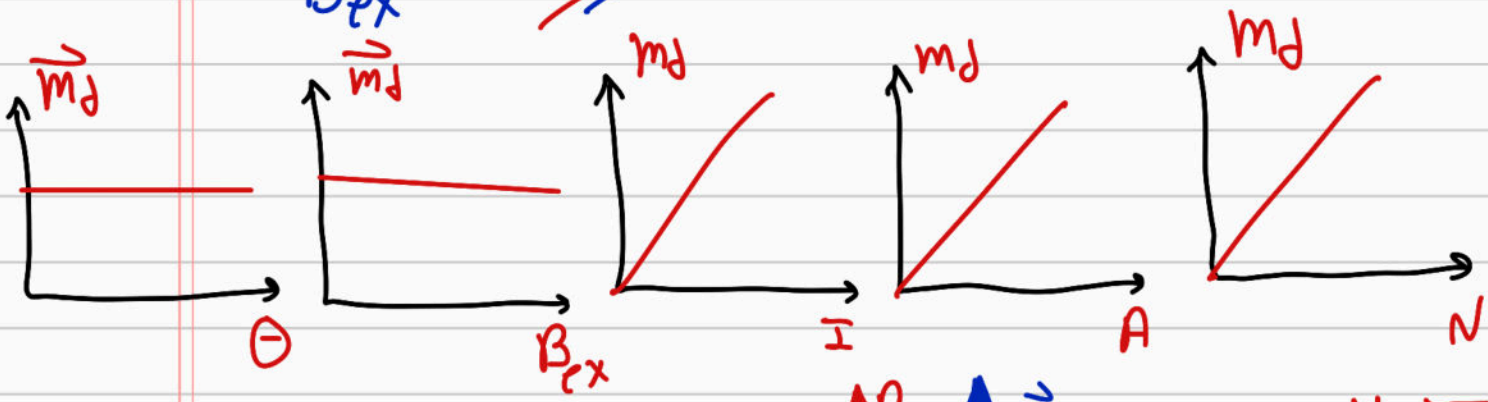
$\tau = B_{ex} I A N \cos\theta$ ✓

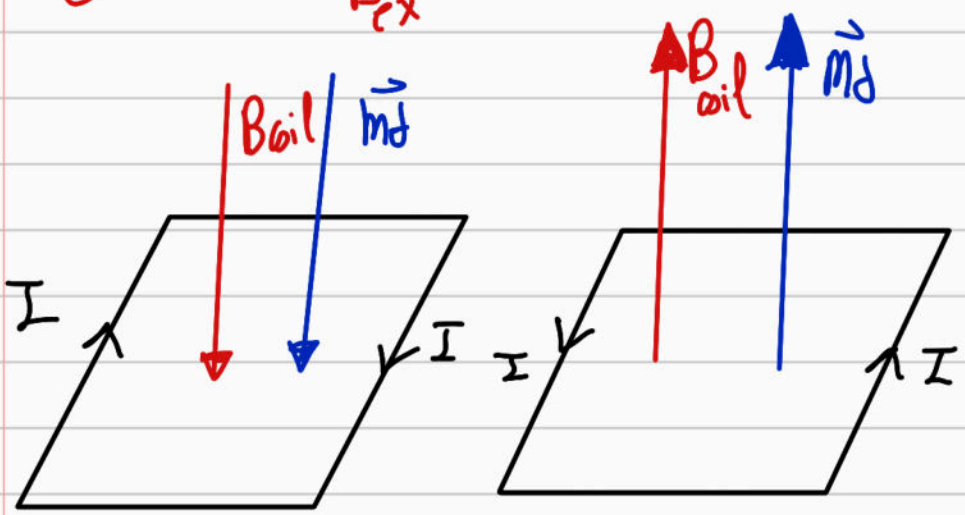
$\tau = B_{ex} I A N \sin\theta$ ✓

magnetic dipole moment ($|\vec{m}_d|$)

Normal Vector acts on the coil pointing To the North of the B_{coil} indicates the Torque sTrength.

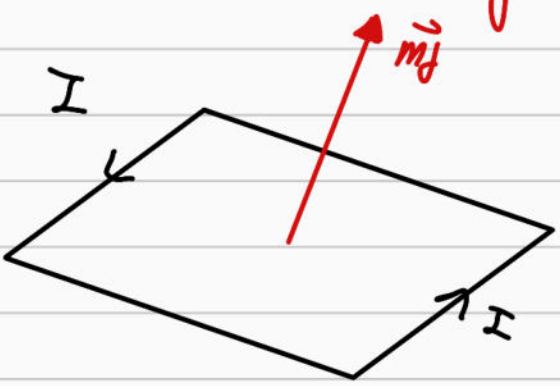
$$|\vec{m}_d| = \frac{\tau_{max}}{B_{ex}} = \frac{B_{ex} IAN}{B_{ex}} = IAN$$



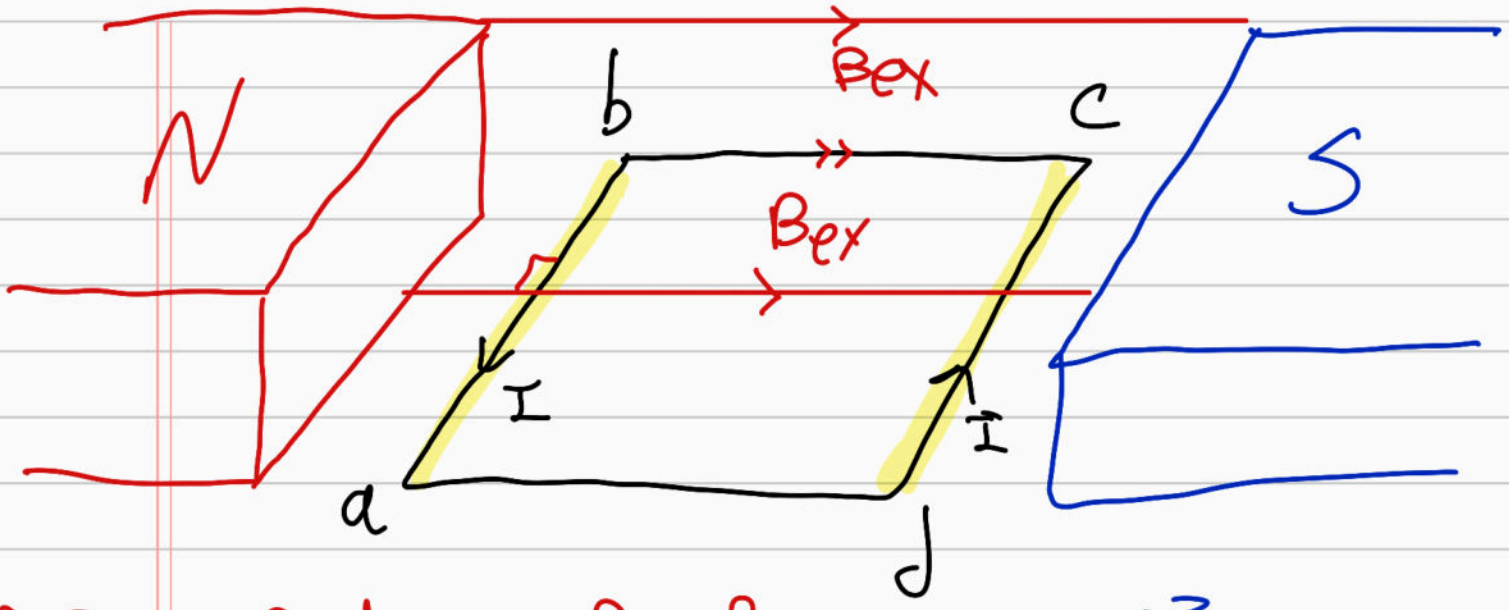


$$B_{coil} = \frac{\mu N I}{2r}$$

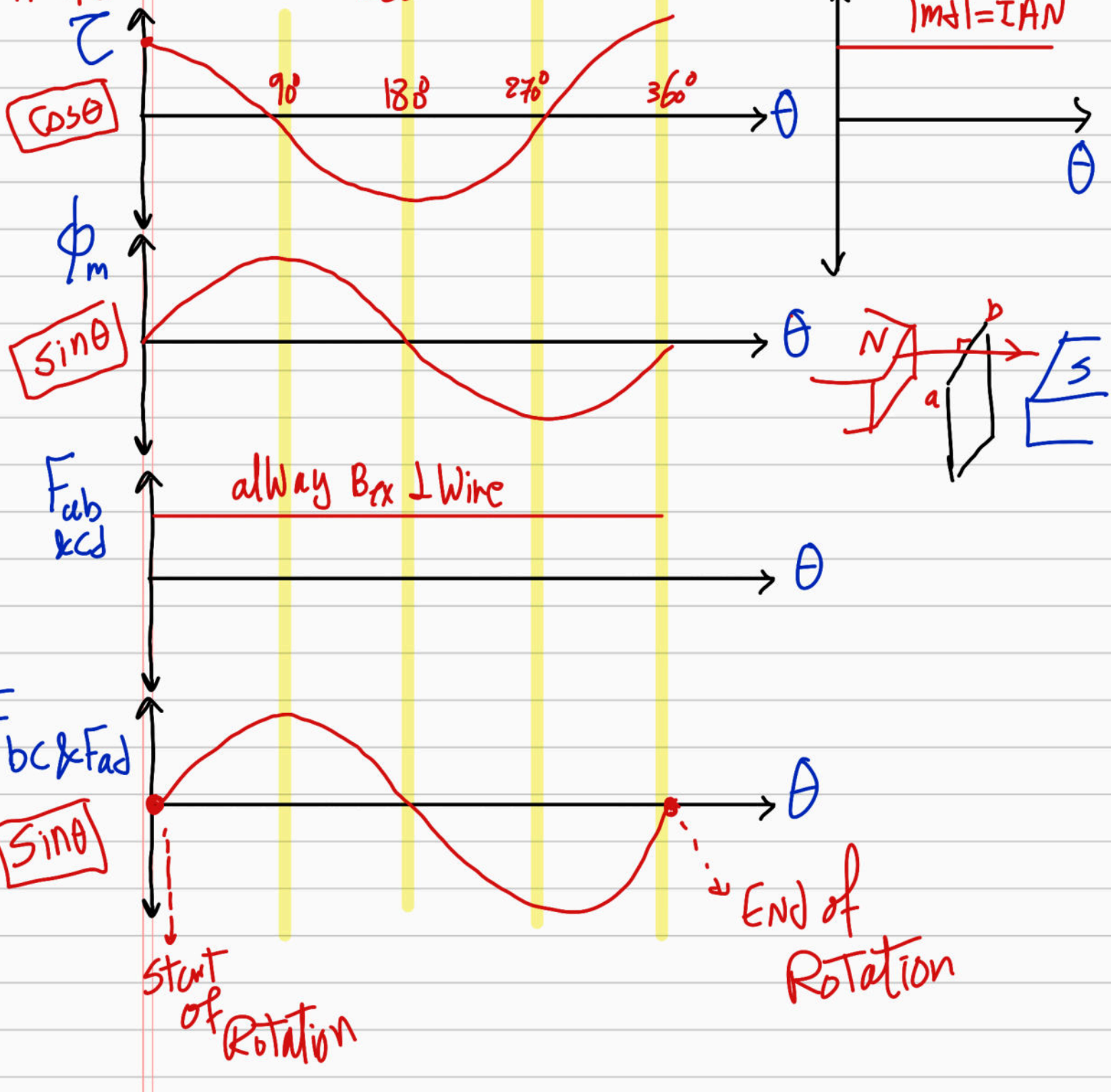
Mag. dipole moment doesn't depend on postion allways Normal To coil plane

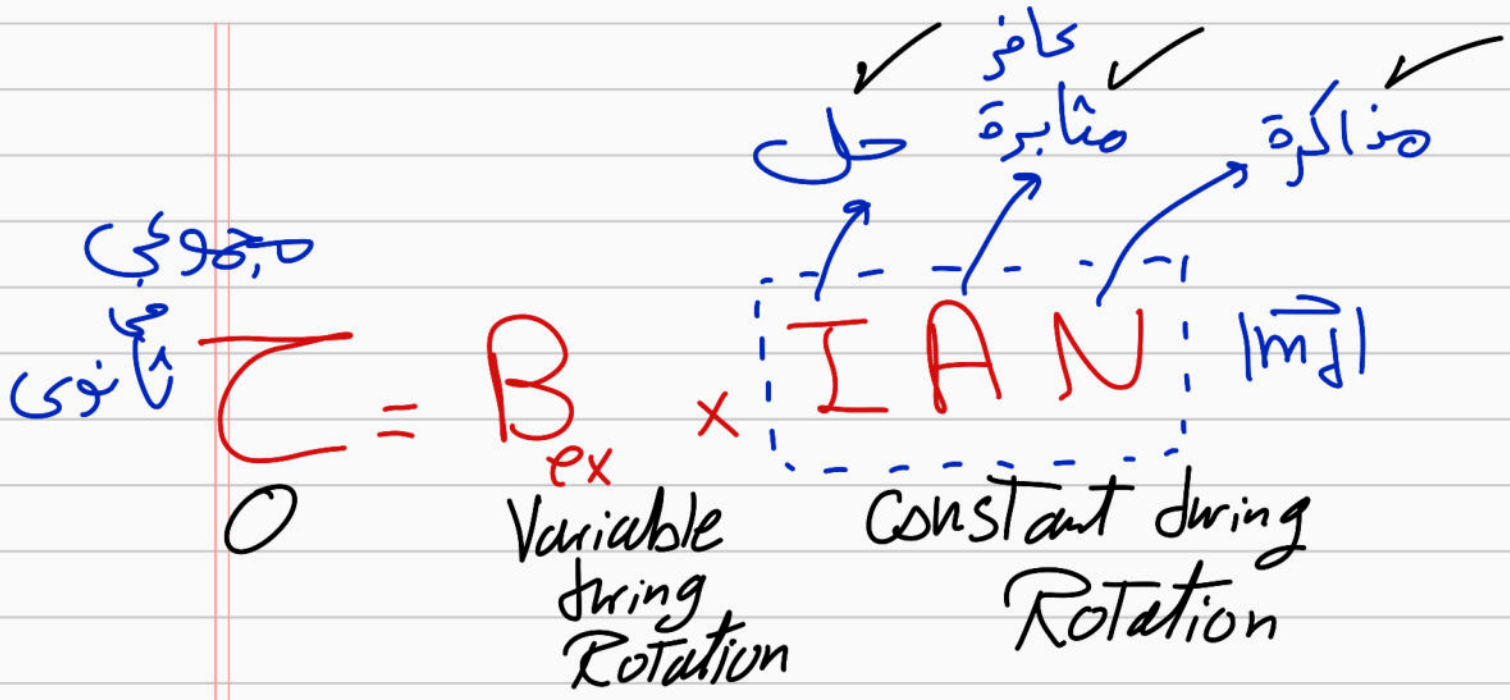






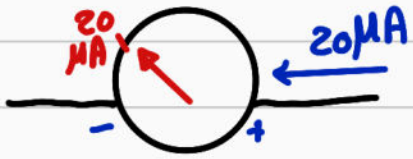
if the coil Rotates 360° from this Position





Position	Φ_m	F	τ	m_d
$B_{ex} \perp$ Coil Wire	$B_{ex} A$	$L I B_{ex}$	Zero	IAN
$B_{ex} \parallel$ Coil Wire	Zero	Zero	$B_{ex} IAN$	IAN
B_{ex} θ Coil Wire	$B_{ex} A \sin\theta$	$L I B_{ex} \sin\theta$	$B_{ex} IAN \cos\theta$	IAN
B_{ex} θ Normal	$B_{ex} A \cos\theta$	$L I B_{ex} \cos\theta$	$B_{ex} IAN \sin\theta$	IAN
Rotate from Perpendicular	$B_{ex} A \cos\theta$ start max	$L I B_{ex} \cos\theta$ start max	$B_{ex} IAN \sin\theta$ start Zero	IAN
Rotate from Parallel	$B_{ex} A \sin\theta$ start Zero	$L I B_{ex} \sin\theta$ start Zero	$B_{ex} IAN \cos\theta$ start max	IAN

Lesson ④

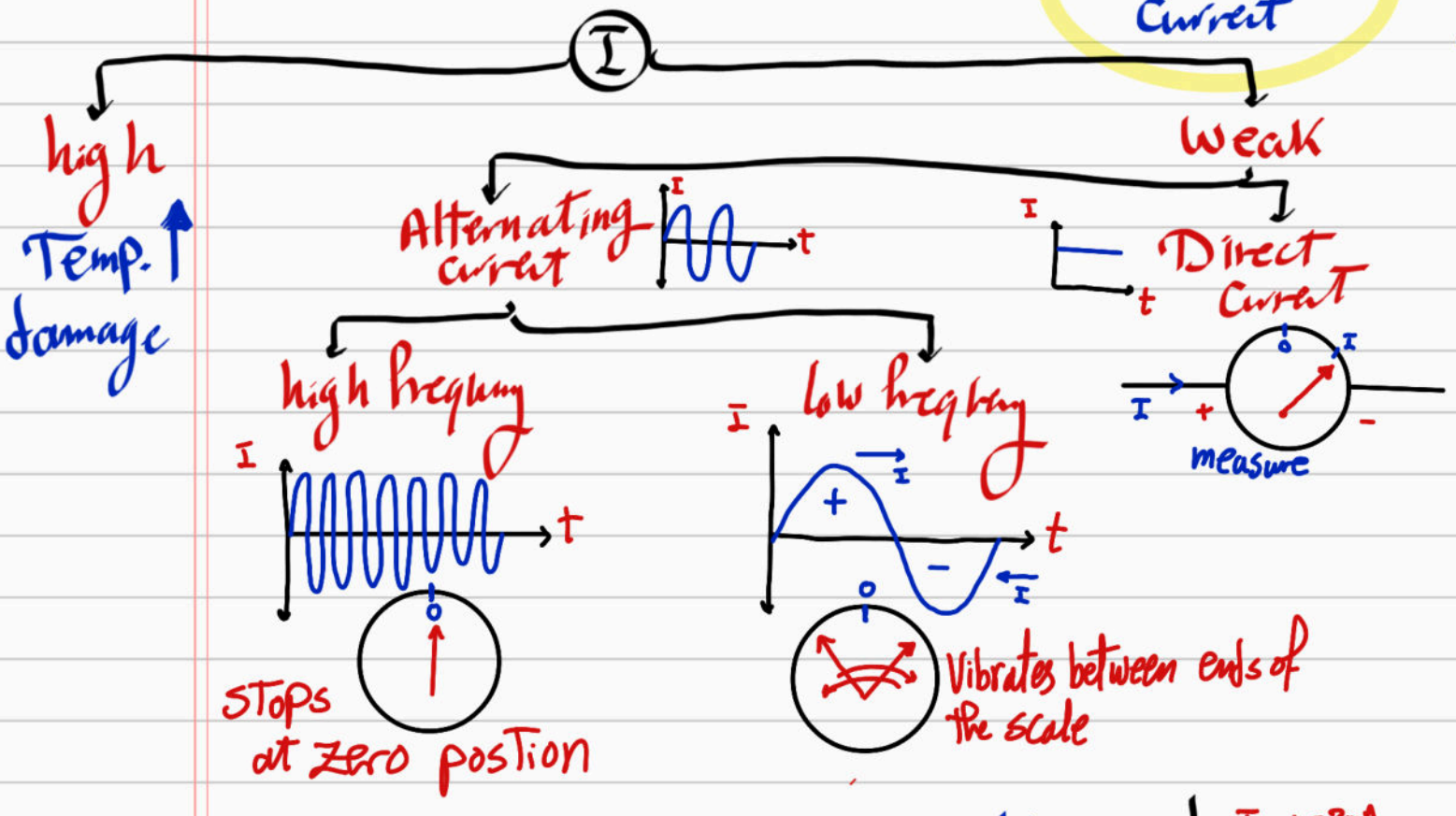


1 Galvanometer:

#Sci. idea: Torque ↝ $\tau = B_{ex} I A N$

#Uses: measure / determine direction / detect

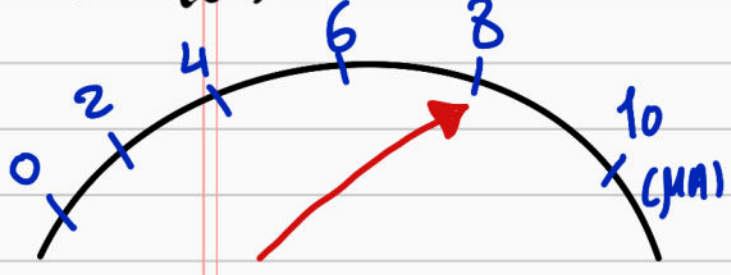
10 mA
20 μA
Weak Direct Current

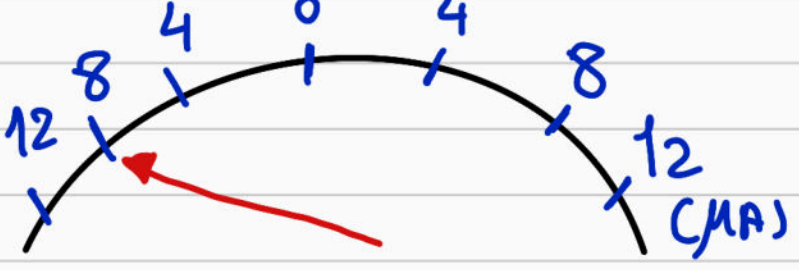


#Connection: Seires [I → constant]



#Scale:



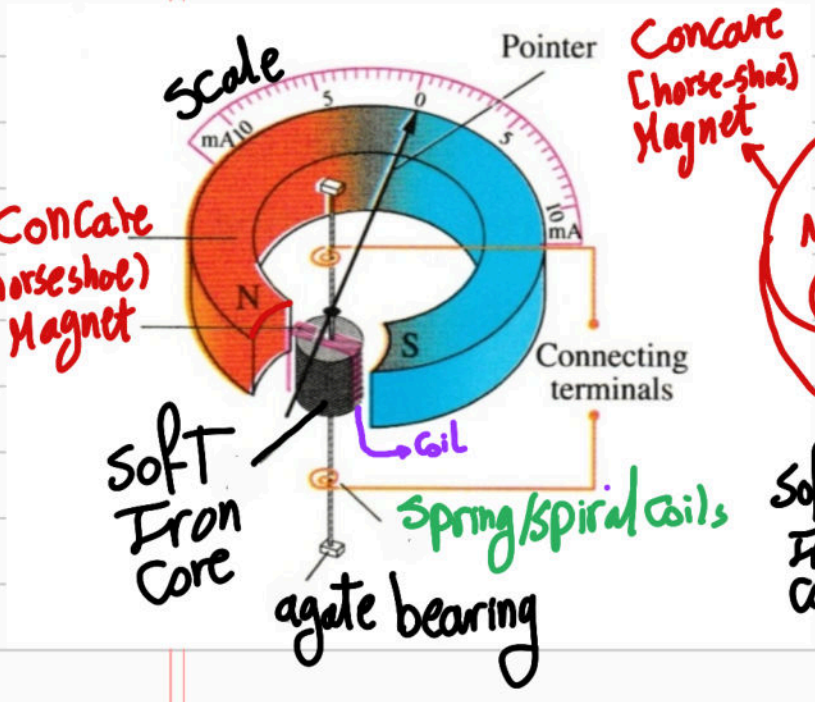


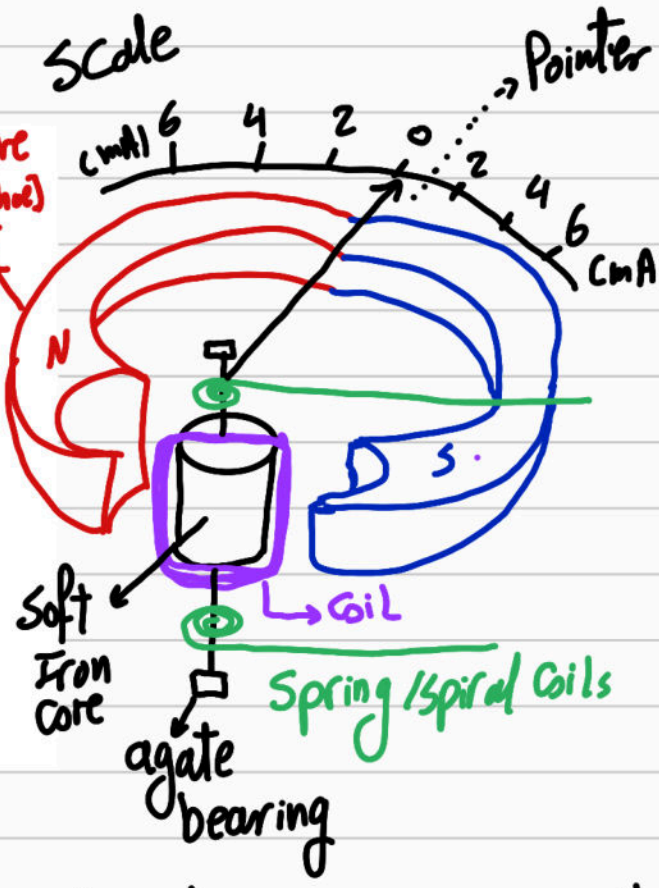
Cannot determine Curent direction

Can determine Curent direction



#Design:

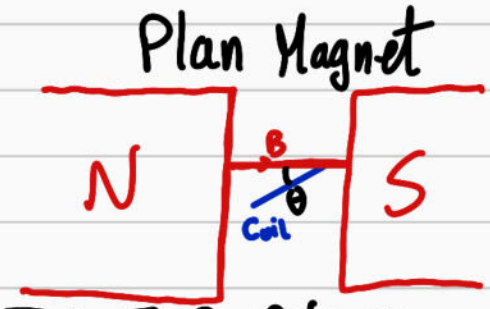




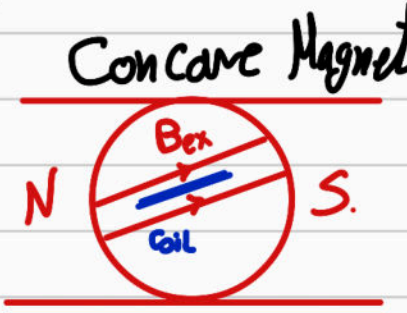
#Components:

1 Concave Magnet:

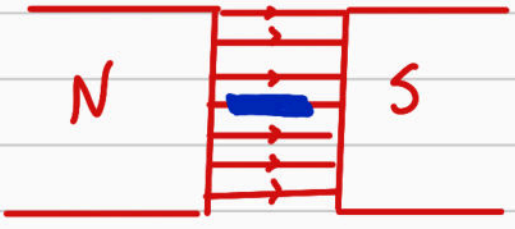
To Concentrate Magnetic flux lines make them Radially so Torque is max always $\tau_{max} = B_{ex} I A N$

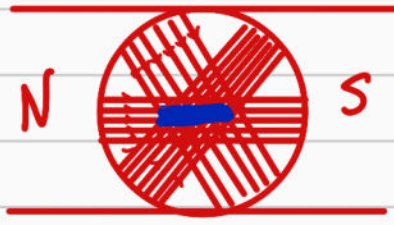


$\tau_{max} > \tau = B_{ex} I A N \cos\theta$



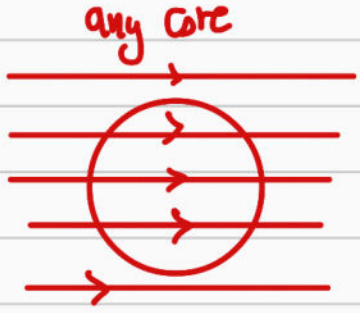
$\tau_{max} = B_{ex} I A N \cos(0)$





2 Soft Iron Core:

due To iTs high premability it helps To concentrate flux To keep The Torque always max



$\mu_{air} = 4\pi \times 10^{-7}$ Tm/A



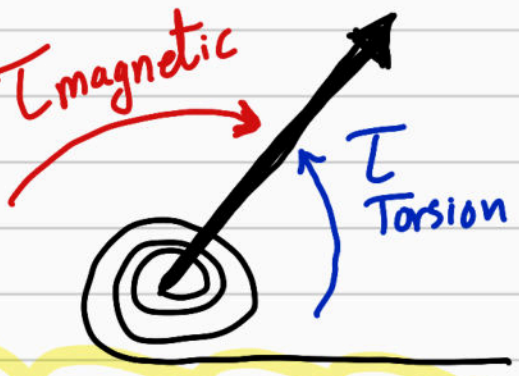
$\mu_{Iron} = 2 \times 10^{-3}$ Tm/A

$\mu_{Iron} \ggg \mu_{air}$

[3] agate bearing: decrease friction force between coil & soft Iron core
(Rotate) (static)

[4] Spiral spring/coils:

Produce Mechanical Torsion [Twisting] To resist the Mag. Torque as the pointer deflects its value increases gradually Till it is equal To the Magnetic Torque So it stops The Pointer at certain Reading.



As the Pointer deflects

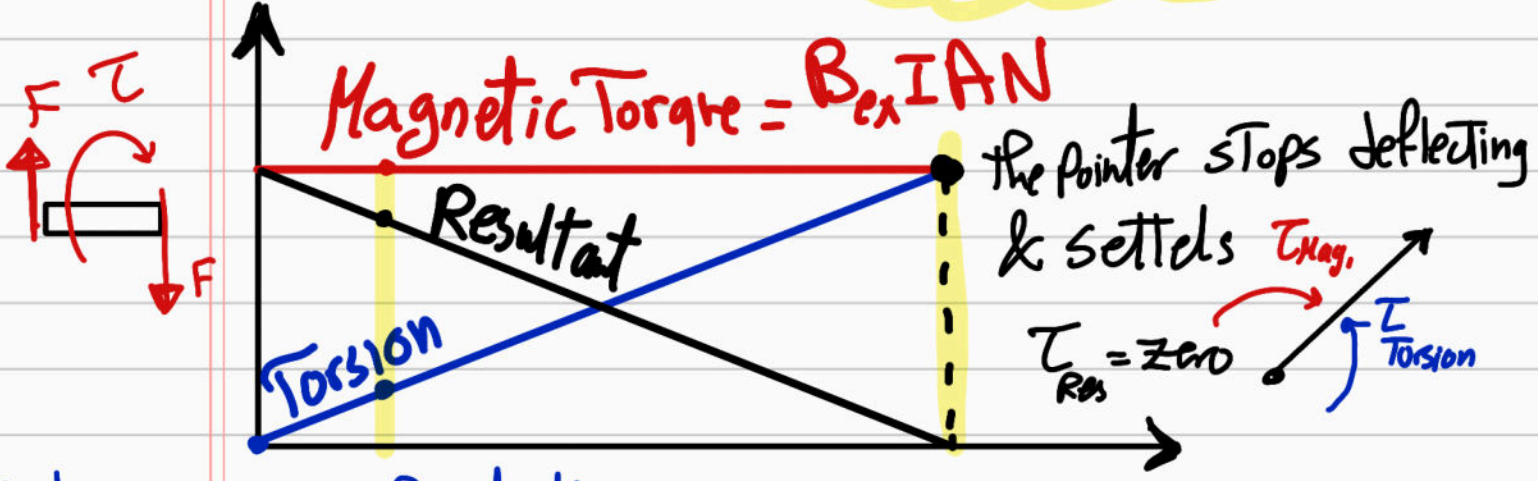
$\tau_{Mag.} > \tau_{Torsion} \rightarrow \tau_{Resultant} > Zero$

When the pointer settels [stops]

$\tau_{Mag.} = \tau_{Torsion} \rightarrow \tau_{Resultant} = Zero$

$\tau_{Resultant} = \tau_{Mag.} - \tau_{Torsion}$

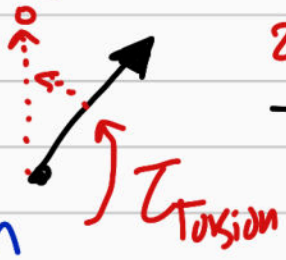
Const.

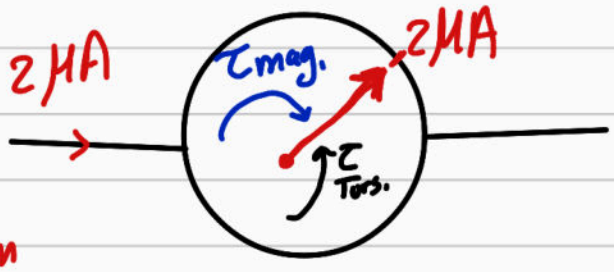


When opening the electric circuit

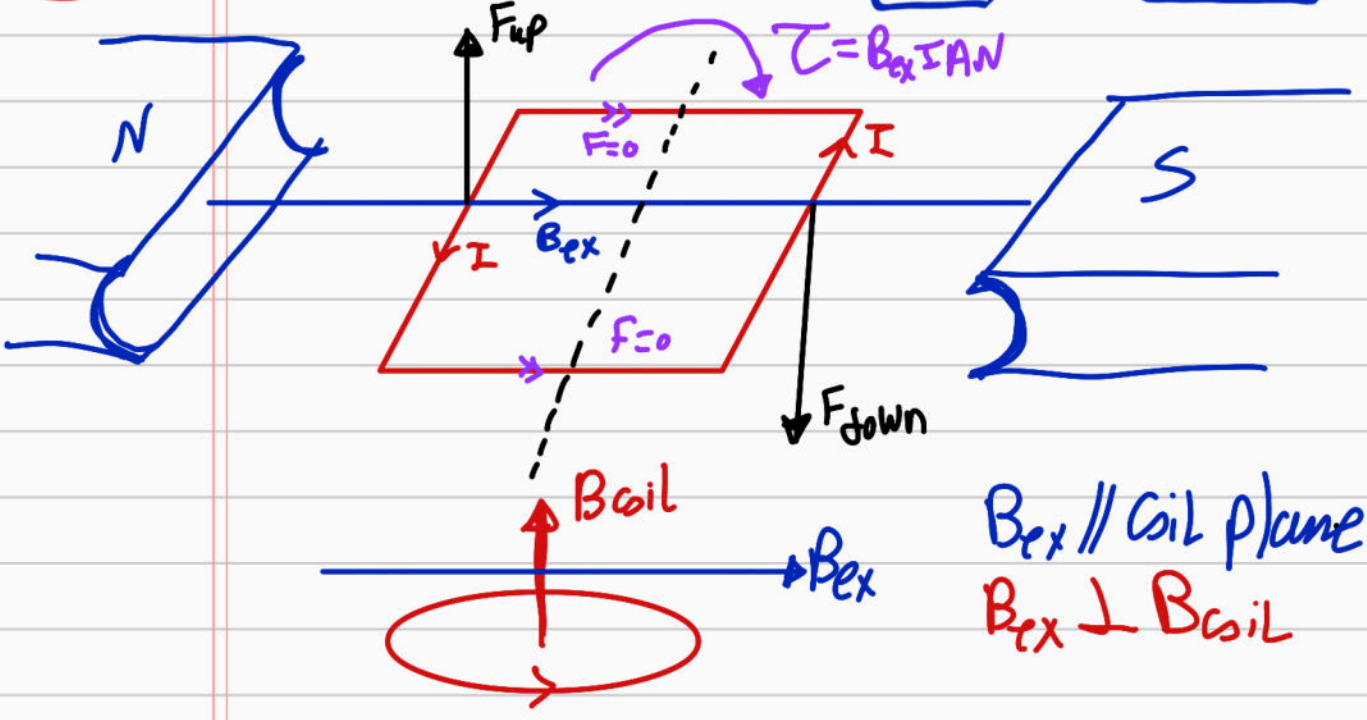
$(I=0) \rightarrow \tau_{mag.} = Zero$

So the Torsion ReTurn Pointer To Zero position



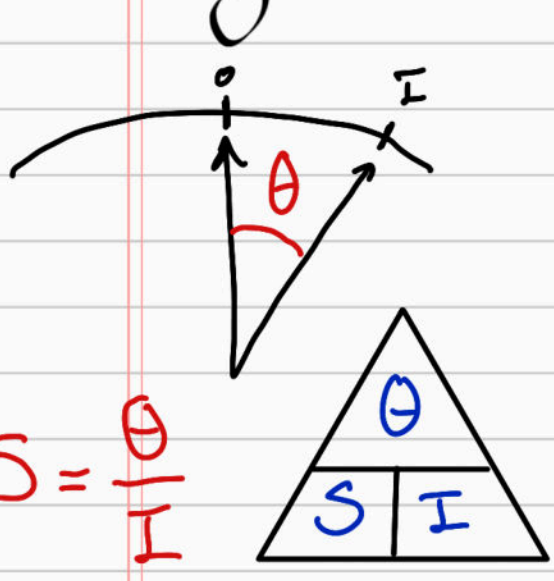


5 Coil → made of Copper | Pe ↓ | I | Temp. ↓



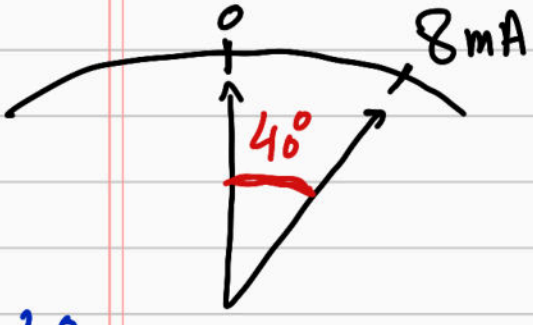
Bex // Coil plane
Bex ⊥ Bcoil

#Sensitivity: scale deflection per unit Current

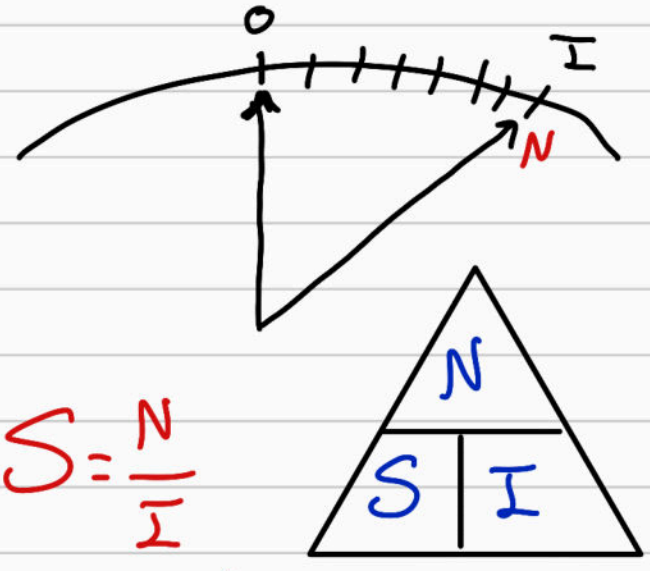


$S = \frac{\theta}{I}$

Unit: degree/mA

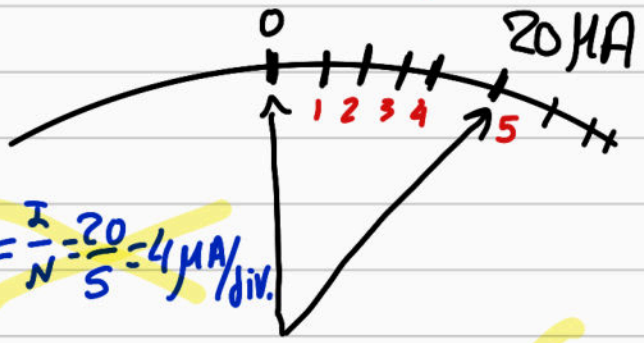


$S = \frac{40°}{8} = 5$ degree/mA



$S = \frac{N}{I}$

Unit: division / μA



$S = \frac{I}{N} = \frac{20}{5} = 4$ μA/div.

$S = \frac{N}{I} = \frac{5}{20} = 0.25$ division/μA

Factors affecting sensitivity of galvanometer:

$$\text{Sensitivity} \propto \frac{B_{ex} A N}{K}$$

B_{ex} → Mag. flux density of Magnet

A → Area of coil

N → no. of Turns of coil

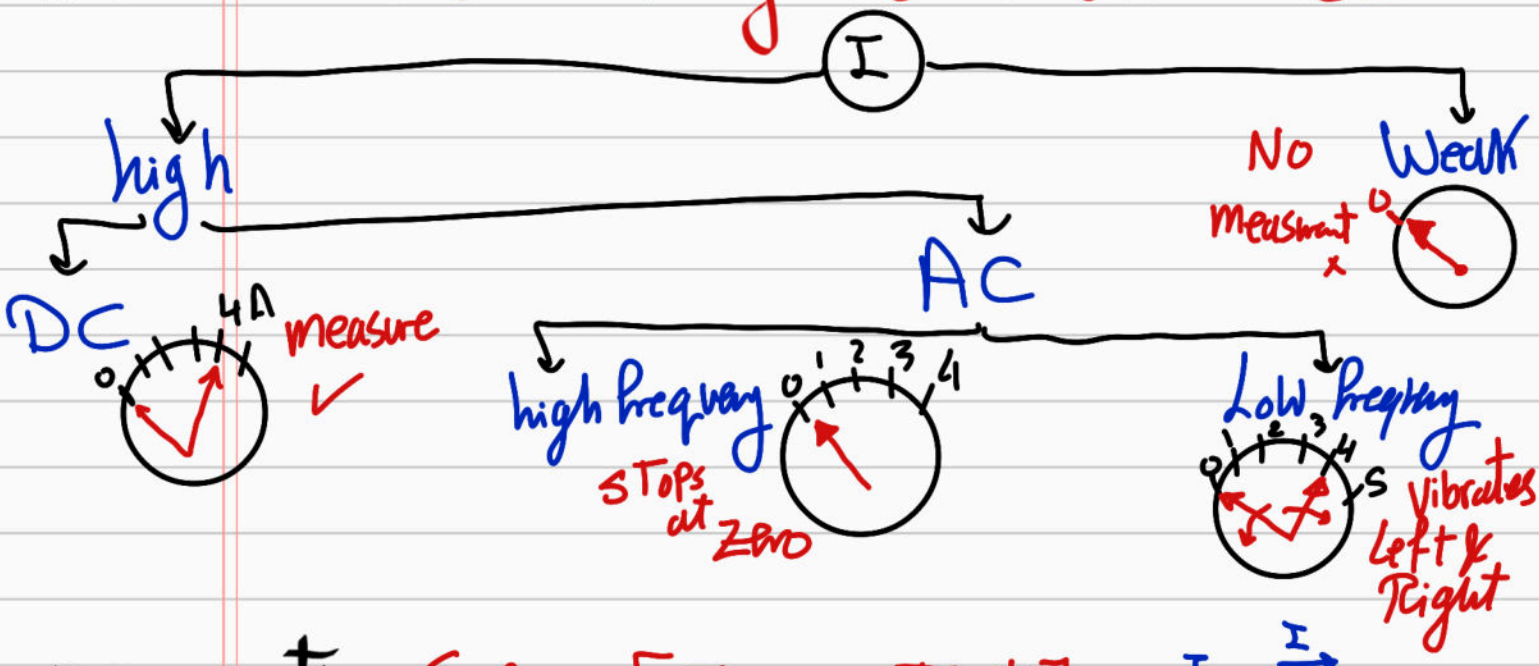
K → Spring Coff.

↑ $S \propto \frac{1}{K}$ ↓ ↓ $K \propto \frac{1}{\text{Elastic}}$ ↑

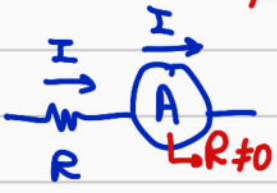
2 Ammeter:

#Sci. idea: Torque

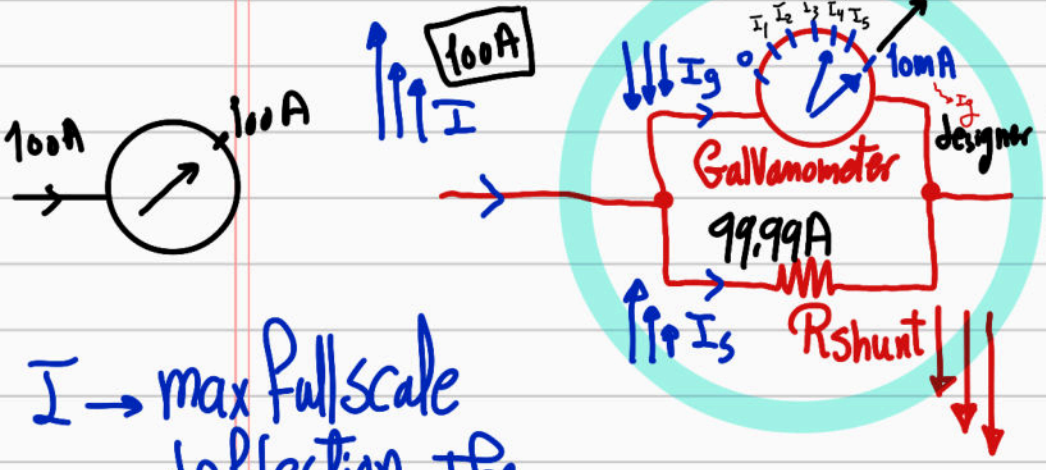
#Uses: measure High Direct current "DC"



#Connection: Series [I → Constant]



design:



I → max Full scale deflection the Ammeter can Withstand.

Ig → max [full scale] def. the galVanometer Can Withstand.

Rg → Resistance of galVanometer

Vg → max [full scale] def. the galVanometer Can Withstand.

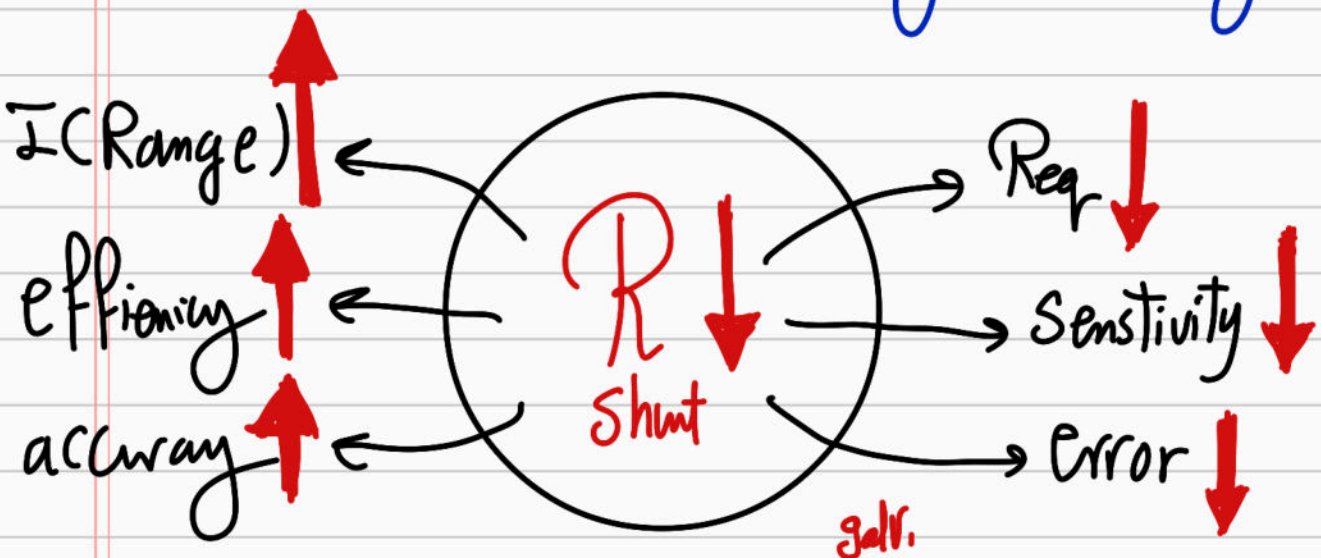
Same galVanometer

Ig → const. Rg → const.

Vg → const.

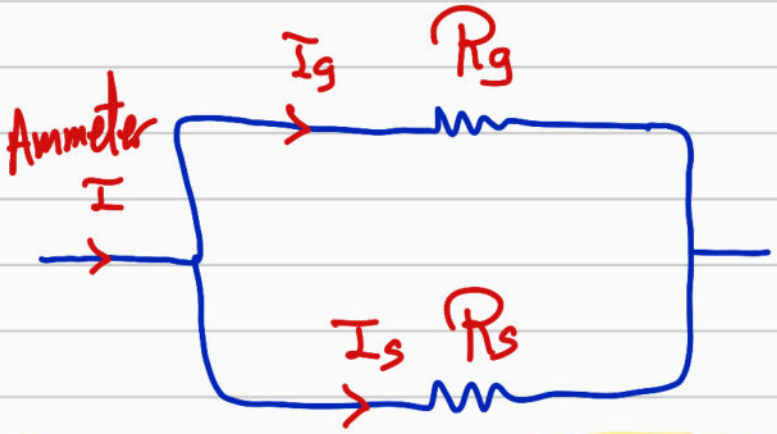
Shunt Resistance:

Small resistance connected in Parallel with the Galvanometer To convert it To Ammeter, by increasing the Measuring Range & decreasing Sensitivity.



Rules:

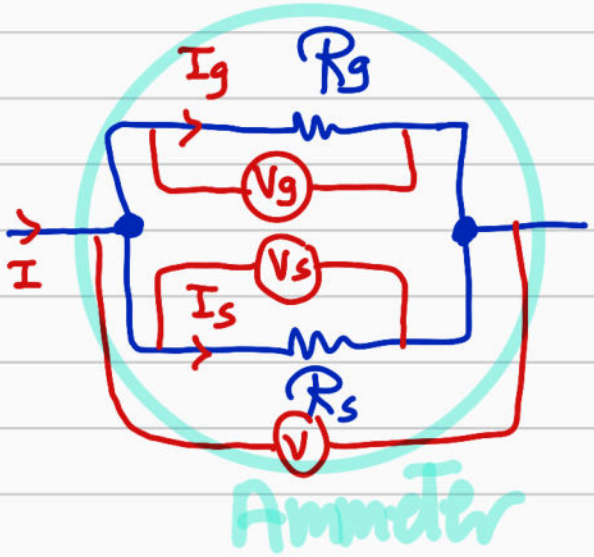
$$R_S = \frac{I_g \times R_g}{I - I_g}$$



Ammeter

$$\text{Sensitivity} = \frac{I_g}{I} = \frac{R_{eq}}{R_g} = \frac{\cancel{R_g} \times R_s}{R_g + R_s} = \frac{R_s}{R_s + R_g}$$

#Rules Summary



1. $V_g = I_g R_g$
2. $V_s = I_s R_s$
3. $V = I R_{eq}$
4. $I = I_s + I_g$
5. $R_{eq} = \frac{R_g \times R_s}{R_g + R_s}$
6. $I_g = \frac{I \times R_s}{R_g + R_s}$
7. $R_s = \frac{I_g R_g}{I - I_g} = \frac{V_g}{I_s} = \frac{V_s}{I_s} = R_s$
8. $S = \frac{I_g}{I} = \frac{R_{eq}}{R_g} = \frac{R_s}{R_s + R_g} = \frac{1}{n} = \frac{1}{n \to (4/3)}$

$V_s = V_g$

$I_s R_s = I_g R_g$

$R_s = \frac{I_g R_g}{I_s}$

$R_s = \frac{I_g R_g}{I - I_g}$

Relations

1. $V = V_s = V_g$
2. $I > I_s > I_g$
3. $R_{eq} < R_s < R_g$

$3/4 = \frac{1}{4/3}$

$I = n \times I_g$

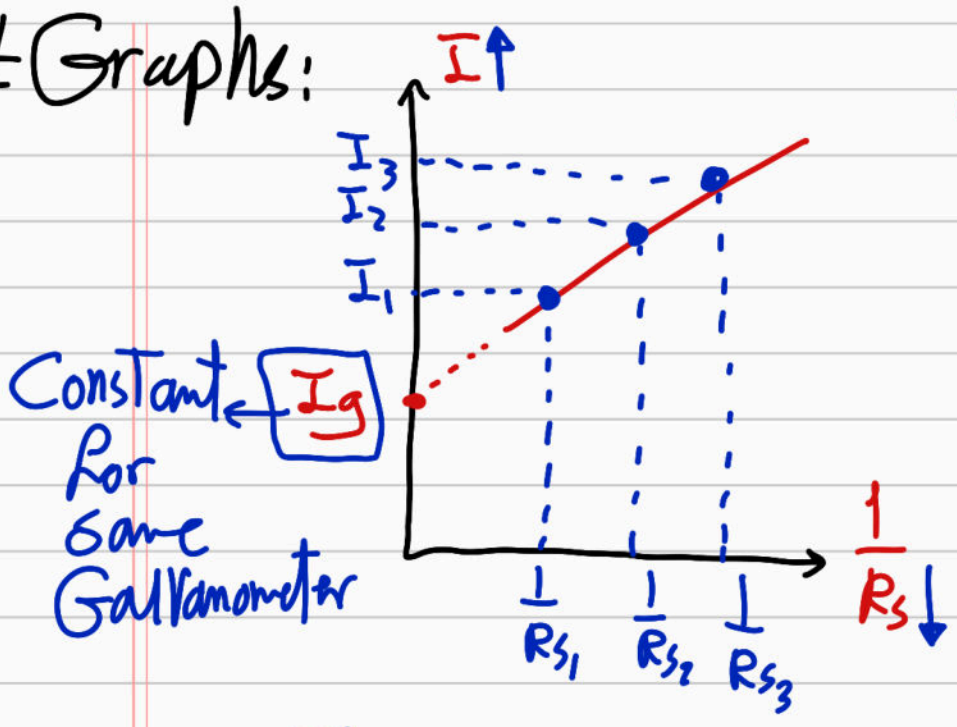
$R_g = n \times R_{eq}$

$R_s + R_g = n \times R_s$

$R_g = n R_s - R_s$

$R_g = (n-1) R_s$

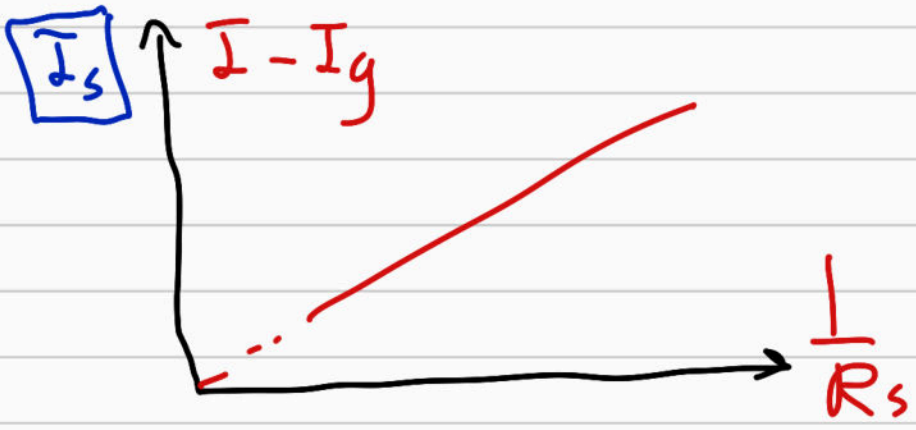
#Graphs:



Ammeter

$R_S \downarrow$ $I(\text{Range}) \uparrow$

I_g, V_g, R_g → Constant



$$\text{Slope} = \frac{I - I_g}{\frac{1}{R_S}} = \frac{I_S}{\frac{1}{R_S}} = I_S R_S = V_S = V = V_g$$

$$R_g = [n-1] R_S$$

$$4\uparrow \quad n = \frac{1}{S\downarrow \frac{1}{4}}$$

$$R_S = \frac{R_g \Rightarrow \text{const.}}{(n-1)} \longrightarrow \boxed{\frac{R_{S1}}{R_{S2}} = \frac{n_2 - 1}{n_1 - 1}}$$

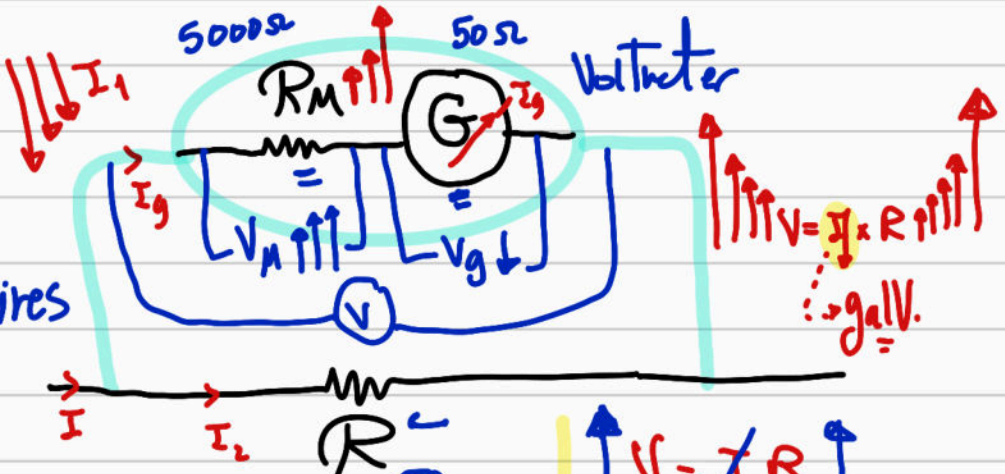
3 Voltmeter:

#Sci. idea: Magnetic Torque

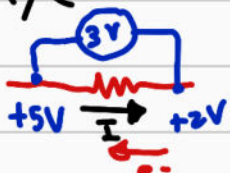
use: measure Potential difference between Two Points

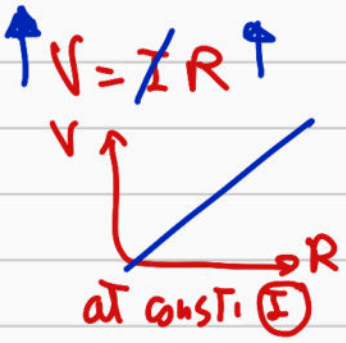
#Connection: Parallel [V → Constant]

#Design:



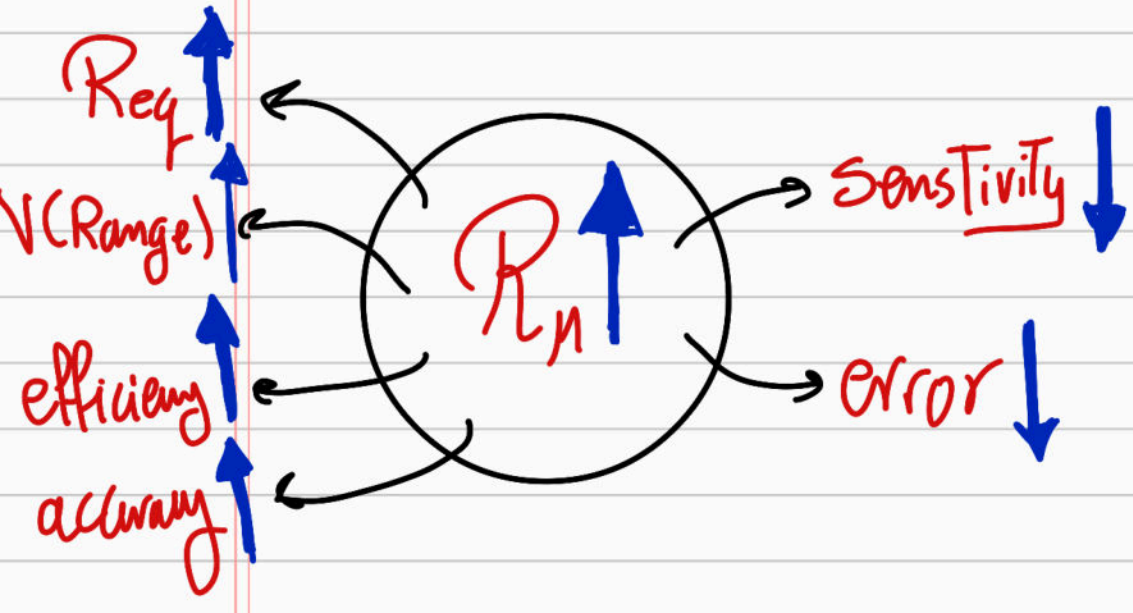
Connected To
- galvanometer → Seires
- Circuit → Parallel

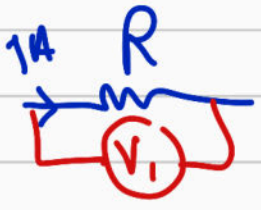


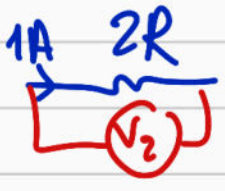


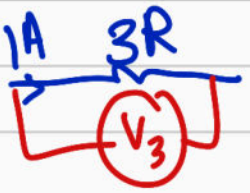
Multiplier Resistance:

Large Resistance Connected in Seires With the galvanometer To Convert it To Voltmeter.

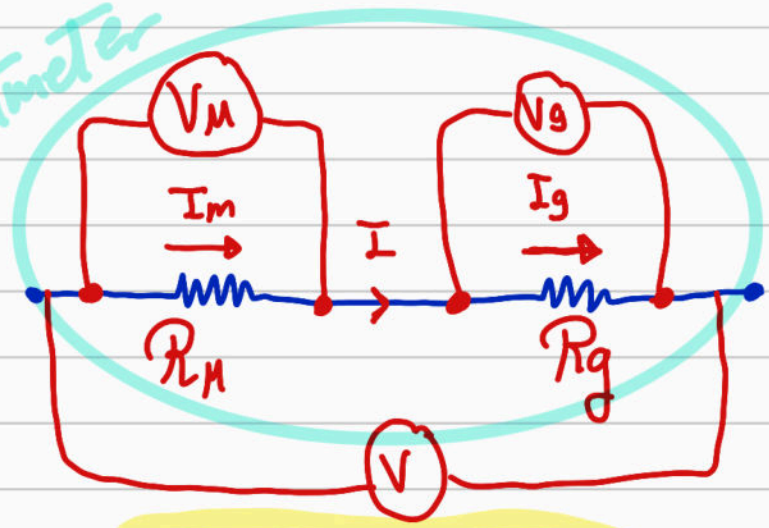








#Rules:



1. $V_g = I_g R_g$
2. $V_M = I_m R_M$
3. $V = I R_{eq}$
4. $V = V_M + V_g$
5. $R_{eq} = R_M + R_g$
6. $R_M = \frac{V - I_g R_g}{I_g} = \frac{V - V_g}{I_g} = \frac{V_M}{I_g} = \frac{V_M}{I_M} = R_M$
7. $S = \frac{V_g}{V} = \frac{R_g}{R_{eq}} = \frac{R_g}{R_g + R_M} = \frac{1}{n}$

Relations

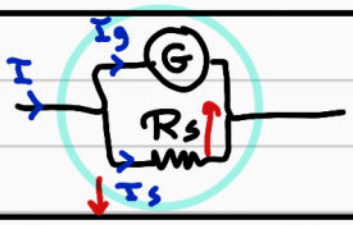
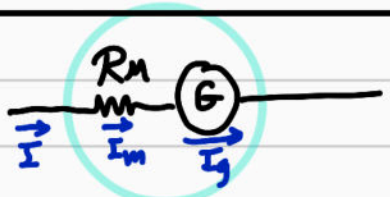
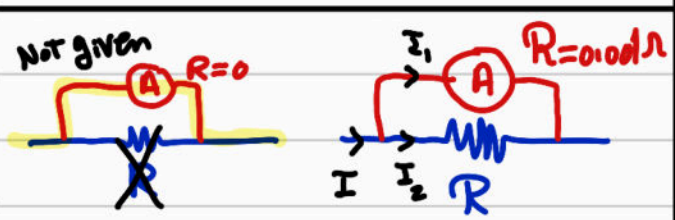
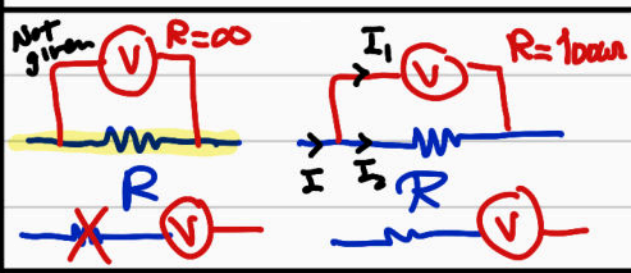
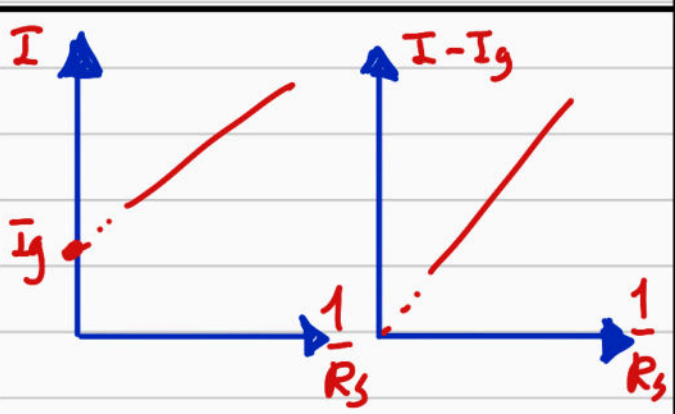
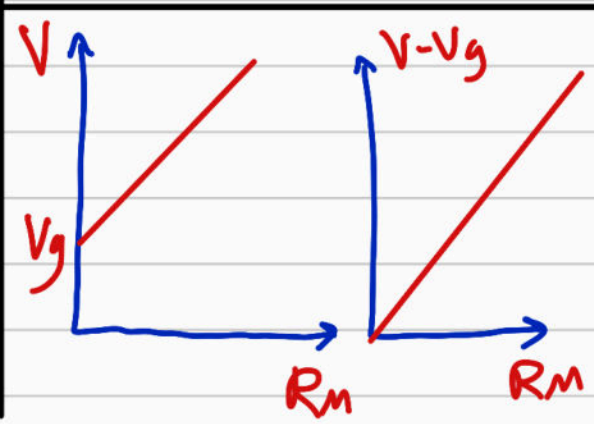
1. $V > V_M > V_g$
2. $R_{eq} > R_M > R_g$
3. $I = I_M = I_g$

Sensitivity dec. To $\frac{1}{n}$ Third $\frac{1}{3} = \frac{1}{n} \rightarrow n = 3$

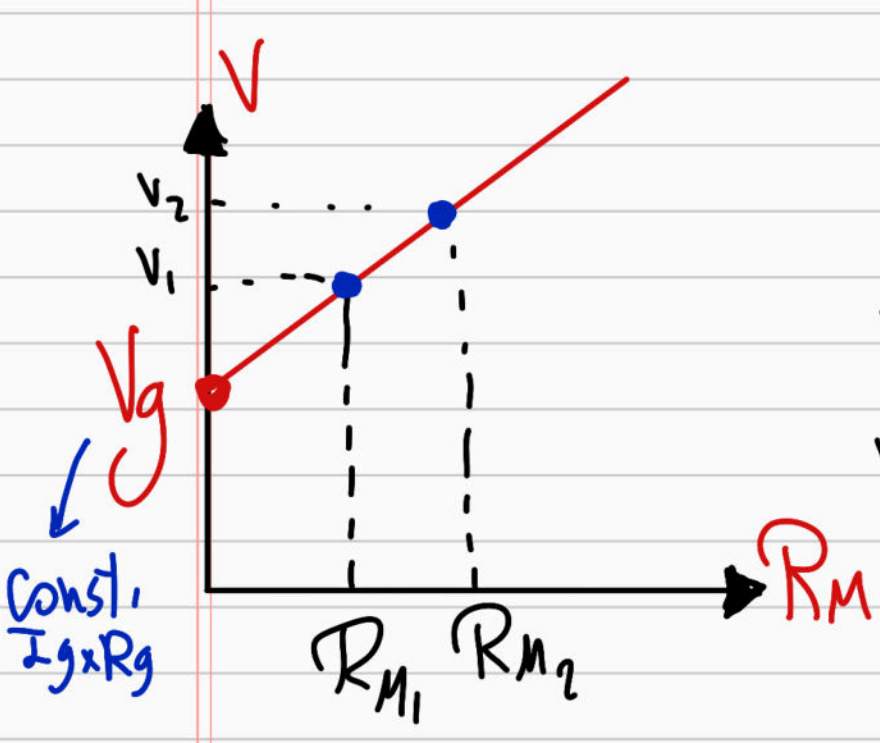
$$\frac{V_g}{V} = \frac{1}{n} \longrightarrow V = n \times V_g$$

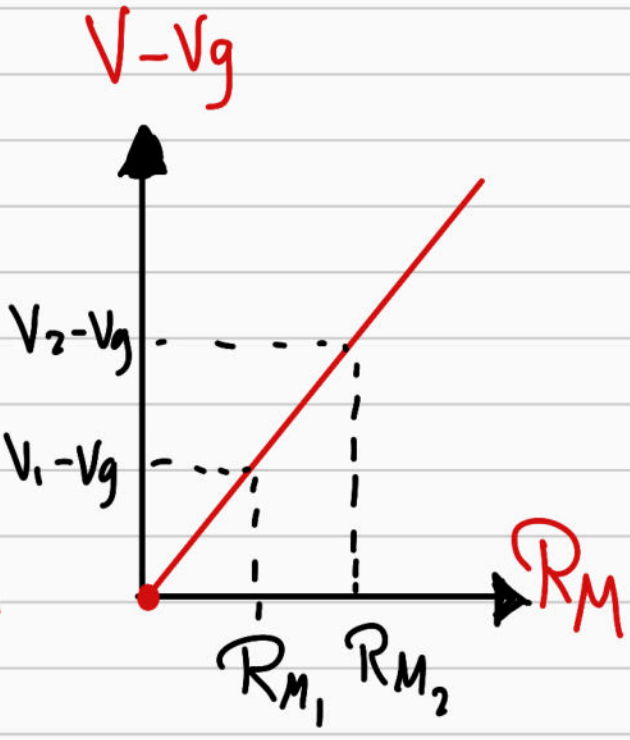
$$\frac{R_g}{R_{eq}} = \frac{1}{n} \longrightarrow R_{eq} = n \times R_g$$

$$\frac{R_g}{R_g + R_M} = \frac{1}{n} \longrightarrow R_M = [n-1] R_g$$

P.o.C	Ammeter	Voltmeter
use	measure High Direct Current	measure Potential difference
connection	Seires [I → const.]	Parallel [V → const.]
Design		
Relations	1. $V = V_s = V_g$ 2. $I > I_s > I_g$ 3. $R_{eq} < R_s < R_g$	1. $I = I_m = I_g$ 2. $V > V_M > V_g$ 3. $R_{eq} > R_M > R_g$
Note ***		
Sensitivity $S \propto \frac{Bex AN}{K}$ spring coff.	$S = \frac{I_g}{I} = \frac{R_{eq}}{R_g} = \frac{R_s}{R_s + R_g} = \frac{1}{n}$ 1) $I = n I_g$ 2) $R_g = n R_{eq}$ 3) $R_g = [n-1] R_s$ $S = \frac{\theta}{I}$ Reading	$S = \frac{V_g}{V} = \frac{R_g}{R_{eq}} = \frac{R_g}{R_g + R_M} = \frac{1}{n}$ 1) $V = n V_g$ 2) $R_{eq} = n R_g$ 3) $R_M = [n-1] R_g$ $S = \frac{\theta}{V}$ Reading
Rule	$R_s = \frac{I_g R_g}{I - I_g}$	$R_M = \frac{V - I_g R_g}{I_g}$
Graphs		

#Graphs:





$$\text{Slope} = \frac{V - V_g}{R_M} = \frac{V_M}{R_M} = I_m = I = I_g$$

#Notes:

(قبل R_m)

1. $V_g \rightarrow$ max / full scale $\rightsquigarrow$ galvanometer can withstand.

2. $V \rightarrow$ max / full scale $\rightsquigarrow$ Voltmeter can withstand.

(بعد الـ R_M)

4 Ohmmeter:

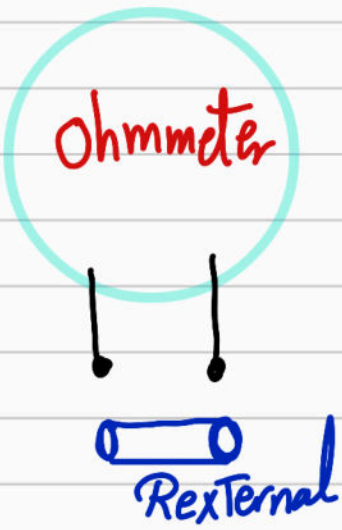
#Sci. Idea: Magnetic Torque [Ohm's Law for closed circuit]

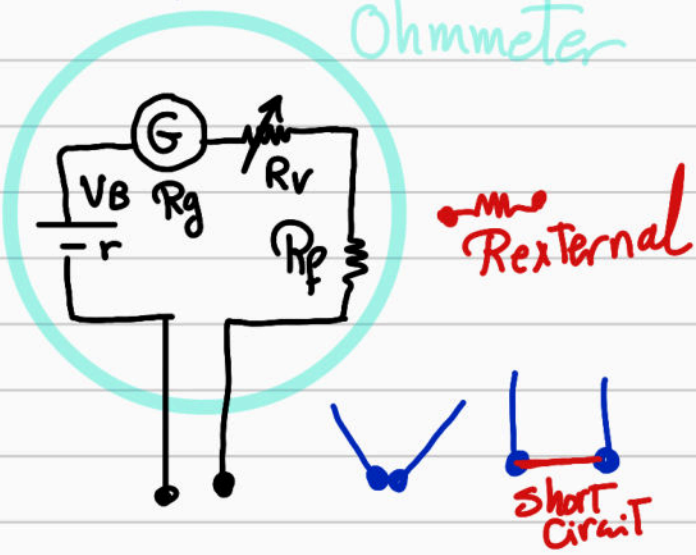
#use: measure Resistance

#Connection: the Resistance is connected Seires To the ohmmeter circuit.

#Design:

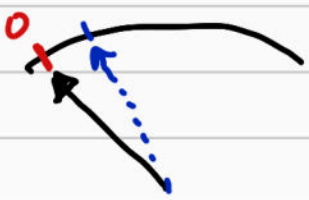
$$I_m = \frac{V_B}{R_{eq}+r}$$





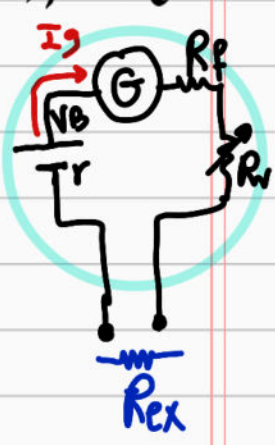
#Standard/fixed/constant Resistance: (Rp): To protect the Galvanometer from high Currat intensity.

#Variable/Rehostate Resistance: (Rv) To adjust/Calibrate the pointer at Zero position.



#Rules:

$$\frac{I_g}{I} = \frac{\frac{V_B}{R_o}}{\frac{V_B}{R_o+R_{ex}}}$$



1. $I_g = \frac{V_B}{R_{ohmmeter}}$ (I_{max})

2. $R_{ohmmeter} = R_g + R_p + R_v + r$ (can be neglected)

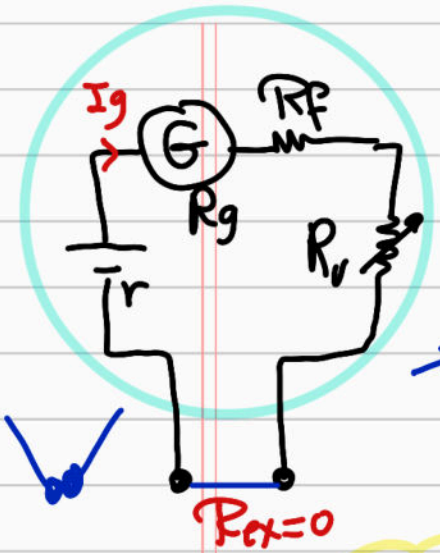
3. $I = \frac{V_B}{R_{ohmmeter} + R_{exTernal}} = \frac{1}{n} I_g$

4. $\frac{I_g}{I} = \frac{R_o + R_{ex}}{R_o} = n$ (ohmmeter, extend)

$$\frac{I}{I_g} = \frac{R_o}{R_o+R_{ex}} = \frac{1}{n}$$

$R_o + R_{ex} = n R_o$

$R_{ex} = n R_o - R_o = [n-1] R_o$



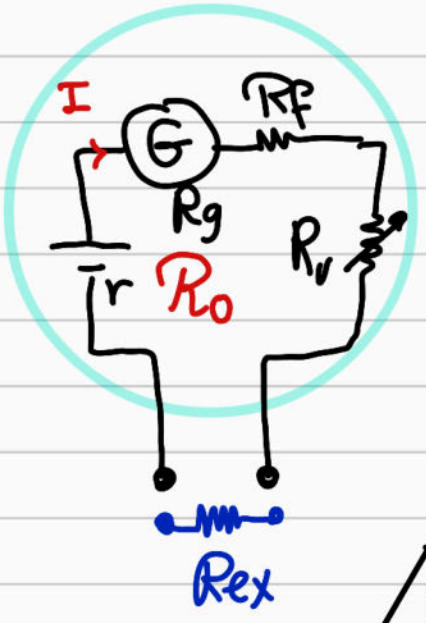
Can be neglected

1 $R_0 = R_g + R_f + R_v + r$

2 $R_0 = \frac{V_B}{I_g}$ → $I_g = \frac{V_B}{R_0}$

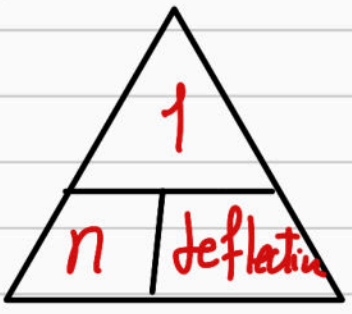
I_g → full scale deflection
→ max Current galv. Can Withstand

$I < I_g$



3 $I = \frac{V_B}{R_0 + R_{ex}} = \frac{1}{n} I_g$

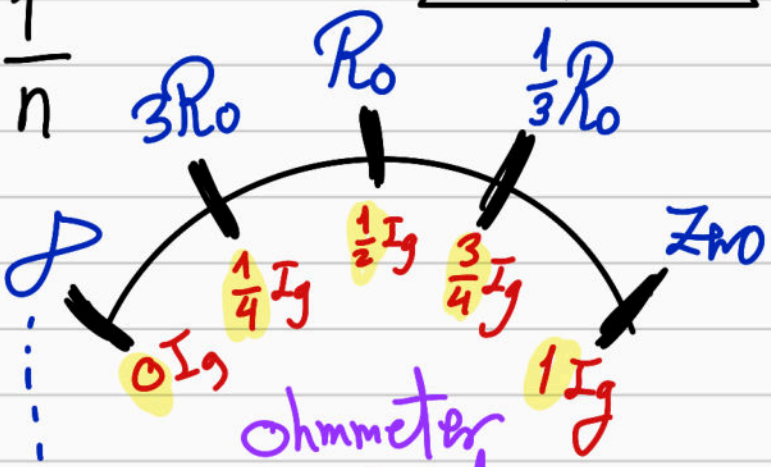
$\frac{I}{I_g} = \frac{\frac{V_B}{R_0 + R_{ex}}}{\frac{V_B}{R_0}} = \frac{R_0}{R_0 + R_{ex}}$



4 $\frac{I}{I_g} = \frac{R_0}{R_0 + R_{ex}} = \frac{1}{n}$

$R_0 + R_{ex} = n R_0$

$R_{ex} = n R_0 - R_0$

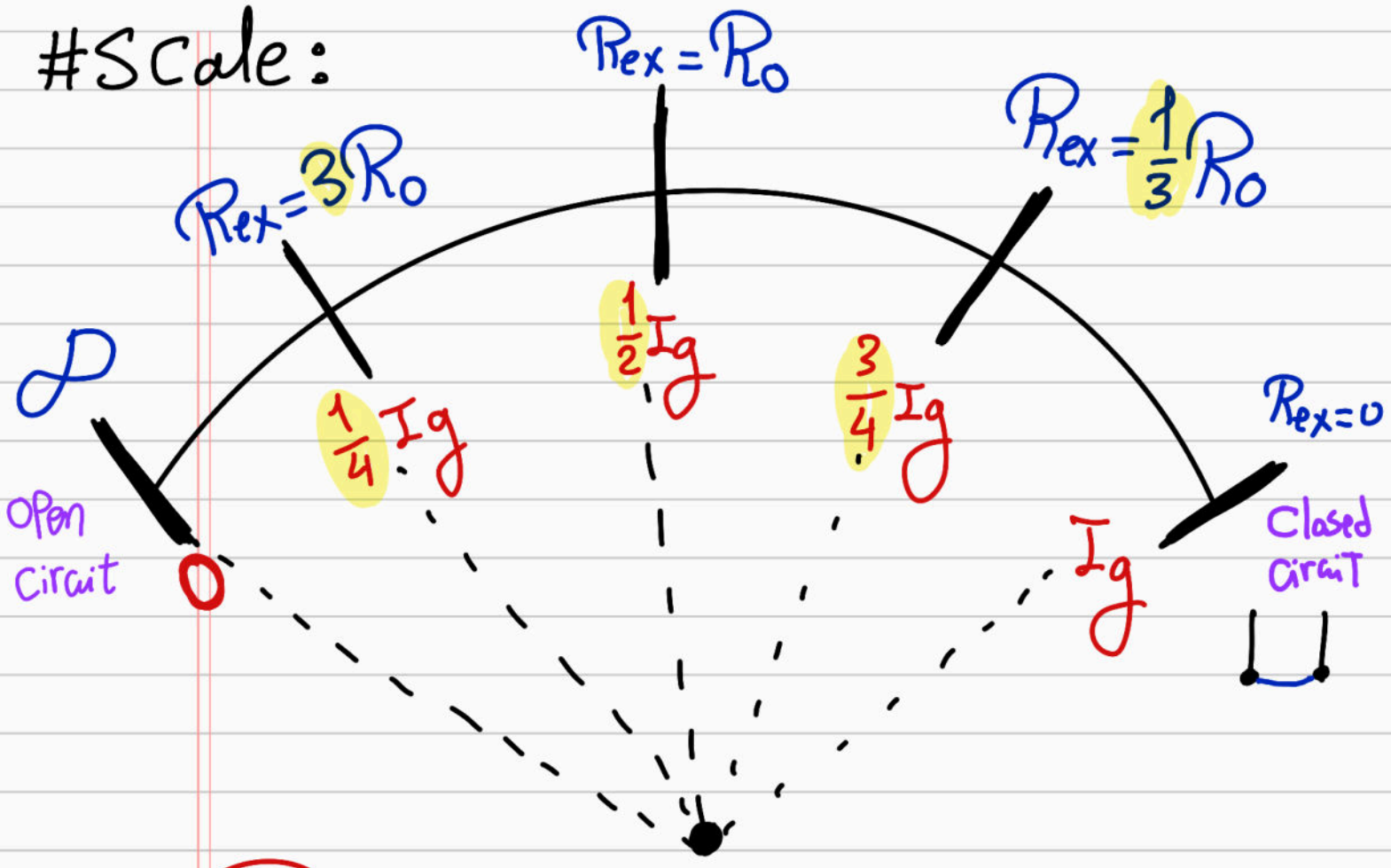


5 $R_{ex} = [n-1] R_0$

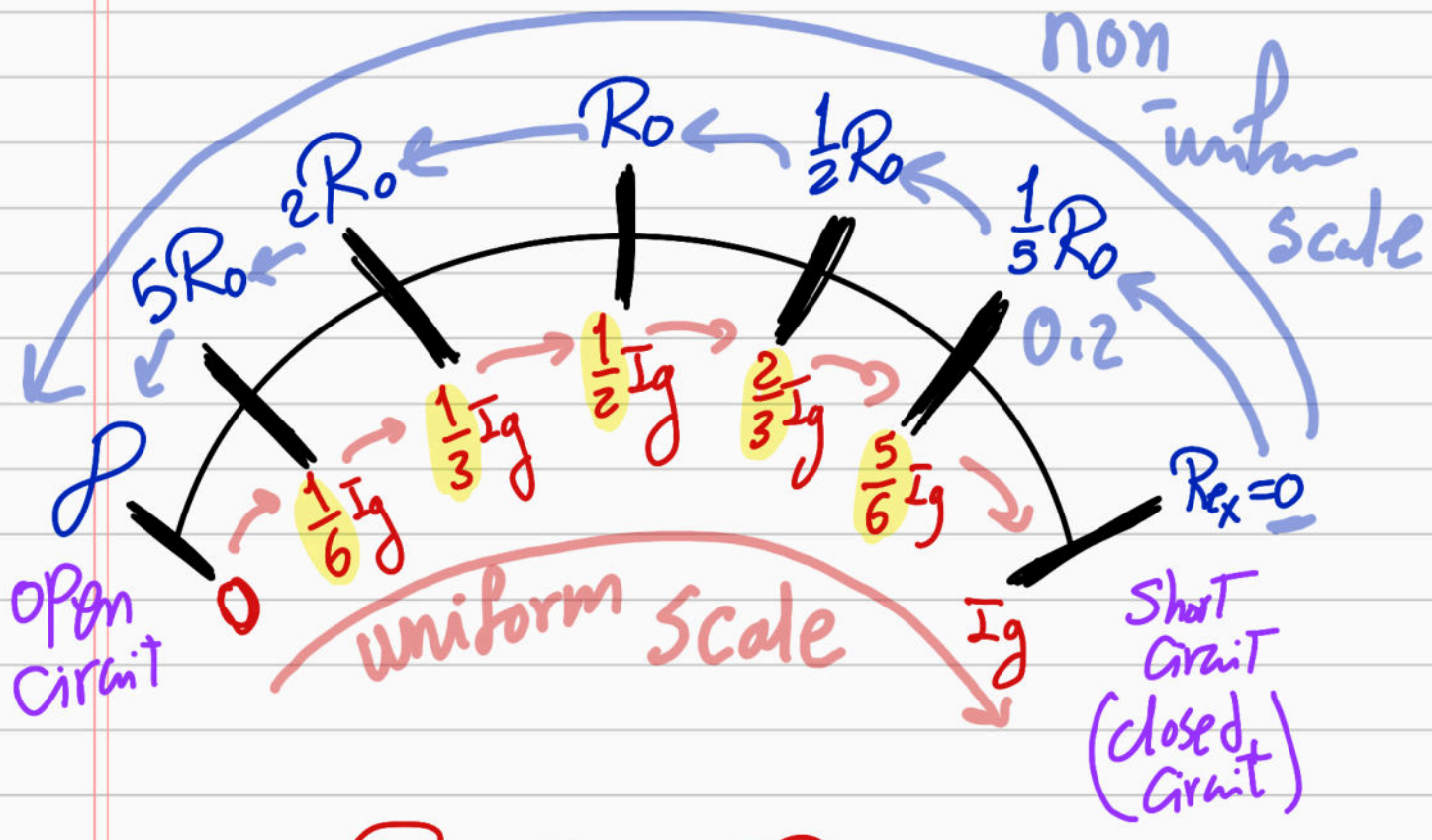
$R_{ex} = [\frac{1}{0} - 1] R_0$

$= \infty$

#Scale:



$$R_{ex} = [n-1]R_0$$



$$R_{ex} = [n-1]R_0$$